Research Article Vol. 33, No. 3 / 10 Feb 2025 / Optics Express 5227

Optics EXPRESS

Reconfigurable intelligent surface-aided positioning-centered multi-LED integrated visible light communication and positioning system

QI WU, JIAN ZHANG,* YAN-YU ZHANG, GANG XIN, AND DUN LI

Information Engineering University, Zhengzhou, Henan Province 450000, China
*Zhang_xinda@126.com

Abstract: This paper investigates a reconfigurable intelligent surface (RIS)-aided integrated visible light communication and positioning (VLCP) system, which aims to reduce sensitivity to line-of-sight (LoS) blockage in visible light and simultaneously achieve capabilities for both visible light communication (VLC) and visible light positioning (VLP). The framework of this system is divided into three parts: positioning, channel state information (CSI) estimation, and data transmission. When the LoS links are present, due to the potential adverse positioning effects RIS caused in this scenario, the RIS only plays a role in assisting the communication. Meanwhile, the positioning process is based on the received signal strength (RSS) information from the LoS transmission. If the LoS links are blocked, the RIS provides added non-line-of-sight (NLoS) positioning signals and achieves subsidiary communications accordingly. The Cramer-Rao lower bound (CRLB) on the positioning result is derived and the CSI estimation is conducted based on the results provided by the positioning schemes to reduce the pilot overhead. The achievable rate is evaluated based on the estimated CSI. The configurations of RIS parameters and the power allocated to positioning and communication are jointly optimized to minimize the CRLB while satisfying constraints on the maximum tolerable outage probability, RIS parameter acquisition, and total transmit power. To address this non-convex optimization problem, we proposed an alternative optimization (AO) algorithm to optimize the RIS assignment matrix and the power coefficient alternatively. We employ the worst-case distribution of the Conditional Value-at-Risk (CVaR) along with successive convex approximation (SCA) techniques to derive feasible solutions of the power coefficient optimization subproblem. Meanwhile, we optimize the RIS assignment matrix based on the principle of maximum received power chosen (MRPC). Simulation results demonstrate the influence of RIS in the positioning process in our proposed VLCP system with or without the presence of LoS links, and meanwhile, the relationship between communication performance and positioning performance is validated through the simulations. From the simulation results, the RIS-aided VLCP system can be verified as a viable option for the upcoming era of sixth-generation (6G) technology.

1. Introduction

In order to alleviate the increasing spectrum crisis faced by radio frequency (RF) communications in the upcoming era of the six generation (6G), with wide and unlicensed bandwidth, the spectrum of visible light is considered to be utilized to complement the RF communication [1]. By modulating the intensity of light emitting diodes (LED), visible light communication (VLC) can achieve both illumination and communication, making it an energy-efficient and environmentally friendly communication technology [2]. Meanwhile, based on the channel characteristics of optical transmission links, visible light positioning (VLP) technology has recently been investigated for its capability to provide high accuracy positioning and is expected to be widely used in indoor scenarios within future 6G ubiquitous networks [3].

#545602 https://doi.org/10.1364/OE.545602
Journal © 2025 Received 22 Oct 2024; revised 5 Dec 2024; accepted 4 Jan 2025; published 3 Feb 2025

In the past, VLC and VLP technologies are developed respectively. On the one hand, according to the research in [4], people spend most of their time in indoor scenarios. Consequently, as a kind of indoor wireless communication technology, VLC draws extensive attention of the corresponding researchers recently and is in the ascendant [5]. Apart from alleviating the spectrum pressure of RF communication, VLC has several extra distinct advantages such as low consumption, high energy efficiency, no RF interference, high secrecy feature and so on. All the distinct benefits provided by VLC, as proposed above, indicate that the VLC is a promising technology for high speed indoor connectivity [6] and is willing to promote the realization of Internet of Everything and Intelligent Twins in 6G Smart Thinking [7].

On the other hand, VLP technology can be achieved by several classical positioning algorithms. The optical signals are detected and analyzed by the photo-detector (PD), and consequently, the positioning information is acquired according to corresponding positioning algorithms based on the optical characteristics. The widely-adopted positioning algorithms in VLP includes time of arrival (TOA) [8], time difference of arrival (TDOA) [9], phase of arrival (POA) [10], phase difference of arrival (PDOA) [11], angle of arrival (AOA) [12], and received signal strength (RSS) [13]. However, the RSS positioning algorithm is most commonly adopted in VLP due to its lower complexity and cost compared to other algorithms that require extremely accurate synchronization. Given the low multi-path effects and low measurement cost, unlike RF positioning technology, VLP systems can achieve high accuracy positioning estimation based on the received power measurement. The high positioning accuracy provided by VLP ensures real-time communication and benefits human-machine interaction as well as the Internet of Everything in future 6G ubiquitous networks.

However, in order to dig out the advantages of the two technologies proposed above, the integrated visible light communication and position (VLCP) system draws the researchers' attention in recent years [14–19]. With high-speed communication and high accuracy positioning accuracy, VLCP technology is well-performed in practical indoor daily scenarios. VLCP is a novel technology integrating the advantages of VLC and VLP and is willing to be widely used in future indoor home network. However, there exist several problems omitted by current research on VLCP. Specially, in current VLCP systems, the data transmission is usually achieved based on the channel state information (CSI) estimation acquired by the positioning results from VLP according to the line-of-sight (LoS) positioning signals. Unfortunately, visible light is highly sensitive to LoS transmission, and as a result, LoS occlusions frequently occur in our daily lives. Some researchers propose the application of two PDs with different orientations to solve the intractable problem [20]. While this solution is promising, there exists a more effective method named reconfigurable intelligent surface (RIS) to address this challenging problem in current research.

As a promising technology to reconfigure the wireless transmission environment, RIS in the optical spectrum has been applied to VLC over the last few years to improve the system performance including channel capacity [21,22], achievable data rate [23], spectrum efficiency [24], energy efficiency [25], security rate [26,27] and so on. From the research above, the system performance indicators in RIS-aided VLC system are only related to the communication performance while neglecting the positioning performance indicators in VLP. The hardware implement of optical RIS is investigated in [28] to verify the feasibility of optical RIS. Meanwhile, the application of optical RIS in VLP is investigated in few works [29–31]. In current researches on the RIS-aided VLP system, the RIS is employed to provide a complement for the positioning process when the LoS links are absent. From the researches above, the parameters of RIS are optimized to increase the positioning performance indicator whereas ignoring the communication performance indicators such as achievable data rate in RIS-aided VLC systems. RIS can play an important role in assisting the VLC or VLP in the absence of LoS links (i.e., the presence of LoS blockage). Consequently, in order to overcome the weakness of visible light in communication

Optics EXPRESS

or positioning (sensitive to LoS transmission), combined the potential technological superiority provided by VLCP, we investigate a RIS-aided integrated VLCP system concentrating on the positioning indicator chosen as Cramer-Rao lower bound (CRLB) in this paper. Through the combination, different from current researches on RIS-aided VLC or VLP, the communication and positioning performance indicators can be joint considered in our RIS-aided VLCP system. To the best of our knowledge, the integrated VLCP system in the presence of RIS is first studied in the literature. Our contributions are summarized as follows:

- We investigate an optical RIS-aided integrated VLCP system with or without the presence of LoS transmission and dig out the relationship between the VLP and VLC in the system.
- The CRLB on the positioning result is derived and adopted as the positioning metric in this paper. The achievable data rate is derived and the outage probability is considered as the communication performance metric.
- The optimization problem is formulated with CRLB as the objective, considering positioning and communication power allocation along with RIS parameters as the optimization variables. The optimization formulation is subject to the communication performance indicator constraint defined by the maximum outage probability that the system should satisfy.
- We figure out the alternative optimization (AO) algorithm to solve the intractable non-convex problem. The worst-case distribution of the Conditional Value-at-Risk (CVaR) along with successive convex approximation (SCA) techniques are employed to derive feasible solutions of power coefficient optimization subproblem, and meanwhile, we propose the maximum received power chosen (MRPC) principle to optimize the RIS assignment matrix.
- We make some simulations to demonstrate the influence of RIS in the positioning process in our proposed VLCP system with or without the presence of LoS links, and meanwhile, the relationship between the communication performance and positioning performance is validated through the simulations. The efficiency of RIS in improving the system performance for integrated VLCP systems is demonstrated through the simulations, particularly when LoS links are blocked.

The rest of this paper is organized as follows. Section 2 introduces the integrated VLCP system model. In Section 3, the positioning measurement (CRLB) and communication performance metric (outage probability) are introduced and derived. Section 4 formulates the optimization problem with the CRLB as the objective, the positioning and communication power allocation and RIS parameters as optimization variables, with the maximum outage probability constraint. Section 5 demonstrates our simulations on the efficiency of RIS in the VLCP system and Section 6 concludes the paper.

Notations: In this paper, bold lowercase and uppercase letters are used to indicate vectors and matrices, respectively, while regular letters represent scalars. Additionally, the notations for the transpose of a matrix, vectorization, differential operator, indicator function and statistical expectation are denoted by $(\cdot)^\top$, $\mathrm{vec}(\cdot)$, $\partial(\cdot)$ (or $\nabla(\cdot)$), $\mathbb{I}(\cdot)$ and $\mathbb{E}[\cdot]$, respectively. The Hadamard product is denoted by $\odot$. The set $\mathbb{R}$ represents the domain of real numbers, while $\mathbb{R}^+$ signifies the domain of non-negative real numbers. Furthermore, $\mathrm{diag}([\boldsymbol{m}_1, \ldots, \boldsymbol{m}_n])$ denotes a block diagonal matrix where the ith main diagonal vector is $\boldsymbol{m}_i$, and $[\boldsymbol{M}]_{i,j}$ or $m_{i,j}$ indicates the element at the i-th row and j-th column of the matrix $\boldsymbol{M}$. Meanwhile, $\oplus$ and $\vee$ denote the Exclusive-OR and OR operations, respectively. $\mathbb{S}^n$ denotes the set of real symmetric matrices of dimension (n). $\mathcal{L} \triangleq \{1, \ldots, N_L\}$ means the set of the LED indexes, and $\mathcal{R} \triangleq \{1, \ldots, N_r\}$ represents the set of the RIS elements indexes. $\mathbf{1}$ and $\mathbf{0}$ represent the all-one matrix and all-zero matrix,

Optics EXPRESS

respectively. Finally, $\geq$ indicates that each element on the left-hand side is greater than or equal to its corresponding element on the right-hand side.

2. Integrated VLCP system model

The section indicates the channel model of the RIS-aided integrated VLCP system. Figure 1 shows an optical VLCP system model with the presence of LoS links. There are N_L non-collinear LEDs in the system to guarantee the capability to determine the 3D position of the receiver ($N_L \geq 3$). In this scenario, the RIS will be activated during the communication process and deactivated during the positioning process, controlled by the RIS controller encapsulated in the transmitter. Moreover, when the LoS links are blocked by other users or non-user blockers, the positioning process can be only achieved by the re-transmitted signals from optical RIS. In this latter scenario, the RIS plays an important role in the positioning process, and accordingly, the CSI will be estimated by the positioning results and the communication process will be achieved according to the estimated CSI.

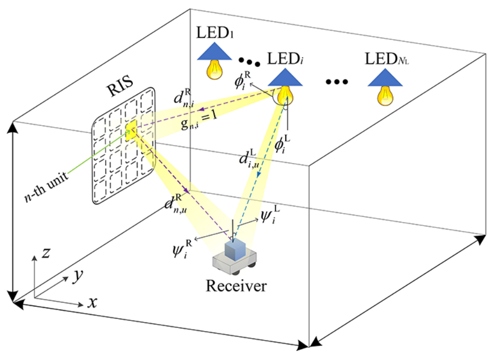


Fig. 1. A RIS-aided integrated VLCP system model.

As shown in Fig. 2, in order to prevent the interference between positioning signals and communication signals, the two types of transmission signals are transmitted using different frequencies. Meanwhile, to confirm the origin of the positioning signals from the corresponding LEDs, we adopt different transmission frequencies for positioning signals emitted by different LEDs. The frequency $f_{p,i}, i \in \mathcal{L}$ assigned to each LED serves as a unique identifier for the i-th LED, enabling differentiation of the positioning signal sources. Consequently, the RSS information can be estimated based on the signals from different LEDs. This structure can be realized using classical frequency-division multiple access (FDMA) in VLC or VLP. Each LED transmits signals encoded with the assigned subcarrier, and the receiver recovers all transmitted signals by performing a discrete Fourier transform (DFT) operation. The receiver is able to process the positioning signals to estimate the receiver position based on RSS scheme, and then, the positioning information is transmitted to the transmitter controller by the uplink based on infrared (IR) communications. Finally, the controller will estimate the CSI based on the feedback information and transmit the communication signal through the downlink via visible light.

In this figure, T_p means the time of transmitting the positioning signals, and meanwhile, T_u and T_c are the time for uplink transmission and downlink communication, respectively. $f_{p,1}, f_{p,2}, \ldots, f_{p,i}, \ldots, f_{p,N_L}$ represent the frequencies assigned to subcarriers reserved for indoor positioning, the subcarrier frequency f_c is with the allocated subcarrier used for broadcasting information, f_u is the frequency of IR communication for the uplink transmission.

Optics EXPRESS

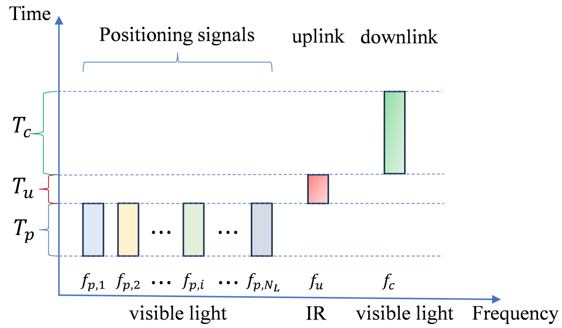


Fig. 2. Frame of the integrated VLCP system.

2.1. Signal and channel model

2.1.1. Integrated communication and positioning signal model

As for the i-th LED, the positioning signal is modulated on the transmission frequency $f_{p,i}$ and the communication signal is modulated on the transmission frequency f_c. Assuming that the communication signal and positioning signal for the i-th LED are denoted by $s_p(t)$ and $s_c(t)$, the integrated communication and positioning signal model for the i-th LED is represented by [32]

$$x_i(t) = \sqrt{p_{p,i}} s_p(t)\cos(2\pi f_{p,i} t) + \sqrt{p_{c,i}} s_c(t)\cos(2\pi f_c t) + I_{DC}, \forall i \in \mathcal{L}, \tag{1}$$

where $p_{p,i}$ and $p_{c,i}$ denote the allocated factor between the positioning and communication, I_{DC} denotes the direct currency (DC) bias caused by intensity-modulation direct-detection (IM/DD) in visible light transmission, $\mathcal{L} \triangleq \{1, \ldots, N_L\}$ denotes the set of the LED index. Without loss of generality, the positioning signal and communication signal are assumed to be satisfied with the corresponding constraints $|s_p(t)| \leq A$, $\mathbb{E}\left\{s_p(t)\right\} = 0$, $\mathbb{E}\left\{s_p^2(t)\right\} = 1$, $|s_c(t)| \leq A$, $\mathbb{E}\left\{s_c(t)\right\} = 0$, $\mathbb{E}\left\{s_c^2(t)\right\} = 1$, where A denotes the constraint on electrical signal and $A>0$.

The DC component can be process via a direct isolating circuit, and therefore, the received positioning signal from the i-th LED is expressed as

$$y_{p,i}(t) = h_i \sqrt{p_{p,i}} s_p(t)\cos(2\pi f_{p,i} t) + n_{p,i}(t), \quad \forall i \in \mathcal{L}, \tag{2}$$

where h_i is the channel gain from the i-th LED to the receiver, $n_{p,i}(t)$ represents the additive white Gaussian noise (AWGN) and $n_{p,i}(t) \sim \mathcal{N}(0, B_p N_0)$, where B_p is the cut-off frequency of the positioning signal and N_0 denotes the noise power.

Since the communication signal is transmitted via the same frequency for different LEDs, the received communication signal in the receiver can be characterized by

$$y_c(t) = \sum_{i=1}^{N_L} y_{c,i}(t) = \mathbf{h}^\top \mathbf{x}_c(t) + n_c(t), \tag{3}$$

where $\mathbf{h} = [h_1, h_2, \ldots, h_i, \ldots, h_{N_L}]^\top$ is the channel matrix of the VLCP system with multi-LED, $\mathbf{x}_c(t)$ denotes the communication signal vector from LEDs and $\mathbf{x}_c(t) = \mathbf{w}_c s_c(t)$, $\mathbf{w}_c$ is the communication power division coefficient vector of different LEDs and $\mathbf{w}_c = [\sqrt{p_{c,1}}, \sqrt{p_{c,2}}, \ldots, \sqrt{p_{c,i}}, \ldots, \sqrt{p_{c,N_L}}]^\top$, $n_c(t)$ represents AWGN and $n_c(t) \sim \mathcal{N}(0, B_c N_0)$, where B_c is the bandwidth of the communication signal.

Considering the constraints on transmitted signals in VLCP systems, the transmitted signal should be subject to the constraints:

$$0 \leq \sqrt{p_{p,i}} + \sqrt{p_{c,i}} \leq \frac{I_{DC}}{A}, \tag{4a}$$

$$0 \leq \sqrt{p_{p,i}} + \sqrt{p_{c,i}} \leq \frac{P_o - I_{DC}}{A}, \tag{4b}$$

$$0 \leq p_{p,i} + p_{c,i} \leq P_e - I_{DC}^2, \quad \forall i \in \mathcal{L}, \tag{4c}$$

where P_o denotes the maximum optical power of each LED, P_e is the maximum electrical power of each LED. The constraint in (4a) guarantees the non-negativity on the transmitted signal in VLCP system, (4b) denotes the peak-power constraint and (4c) means the average-power constraint to confront with the human eye safety requirement and the practical circuit limitations, respectively.

2.1.2. Channel model

In order to emulate the real-life scenario, the LoS blockage problem should be considered in our proposed channel model concluding the user's self-blockage and non-user blockage. Consequently, the channel gain between the i-th LED and the receiver is expressed as

$$h_i = \mathbb{I}_i^{\text{LoS}} h_i^{\text{LoS}} + \mathbb{I}_i^{\text{NLoS}} h_i^{\text{NLoS}}, \quad \forall i \in \mathcal{L}, \tag{5}$$

where $\mathbb{I}_i^{\text{LoS}} \in \{0, 1\}$ denotes the binary indicator function to determine whether the i-th LoS link is blocked. Specially, the i-th LoS link is blocked when $\mathbb{I}_i^{\text{LoS}} = 0$ whereas the i-th LoS link is present when $\mathbb{I}_i^{\text{LoS}} = 1$. Similarly, the binary indicator function $\mathbb{I}_i^{\text{NLoS}} \in \{0, 1\}$ is set to determine when the RIS should be online to assist the communication process and avoid deteriorating the positioning process. h_i^{LoS} and h_i^{NLoS} denote the i-th LoS and NLoS channel gains, respectively. Since the weak reflective effect caused by the regular walls can be ignored [33], the NLoS channel gain only considers the component caused by the RIS in our works.

According to the research in [31], the improvement of RIS on the positioning performance depends on the location of receiver. Therefore, the application of RIS does not always lead to enhanced localization performance, especially when the LoS links are present. In order to avoid the negative effect of RIS in positioning process, we set the RIS is off-line in the positioning process in our proposed VLCP system if LoS links are present. When the LoS links are present, the positioning process is based on the RSS information from LoS transmission and the communication process can be realized by LoS and RIS-reflected links. Meanwhile, when the LoS links are blocked (the LoS link from one LED is sentenced to be blocked if the signal at this frequency cannot be received by the receiver), the RIS is employed to provide added NLoS positioning signals and achieve subsidiary communications accordingly. The two binary indicator functions $\mathbb{I}_i^{\text{LoS}}$ and $\mathbb{I}_i^{\text{NLoS}}$ have a logical relationship which can be characterized as

$$\mathbb{I}_i^{\text{LoS}} \oplus \mathbb{I}_i^{\text{NLoS}} = 1, \quad \text{Positioning process}, \tag{6a}$$

$$\mathbb{I}_i^{\text{LoS}} \vee \mathbb{I}_i^{\text{NLoS}} = 1, \quad \text{Communication process}, \tag{6b}$$

where $\oplus$ and $\vee$ denote the Exclusive-OR and OR operations, respectively. (6a) avoids the negative effect provided by RIS in the positioning process when the LoS links are present, and the RIS can provide added NLoS positioning signals when the LoS link is blocked, (6b) ensures that there is still a communication link even if the LoS link is blocked.

Optics EXPRESS

Consequently, the indicator function $\mathbb{I}_i^{\mathrm{NLoS}}$ can be determined by $\mathbb{I}_i^{\mathrm{LoS}}$. We can define the indicator function $\mathbb{I}_i^{\mathrm{LoS}}$ as

$$\mathbb{I}_i^{\mathrm{LoS}} = I_i^{\mathrm{s}} \times \prod_{m=1}^{M} I_i^{\mathrm{mu,m}}, \quad \forall i \in \mathcal{L}, \tag{7}$$

where $I_i^{\mathrm{s}} \in \{0, 1\}$ indicates if the LoS link is blocked by the user itself and $I_i^{\mathrm{mu,m}} = 0$ denotes that the blockage is caused by the m-th mutual interference from the non-user, and $I_i^{\mathrm{mu,m}} = 1$ means the blockage is absent, M denotes the number of blockers from the non-user, and $\prod$ means the product operator.

Specifically, the i-th LoS channel gain can be expressed as [33]

$$h_i^{\mathrm{LoS}} = \frac{(m+1)A_{re}}{2\pi(d_{i,u}^{\mathrm{L}})^2} \cos^m(\phi_i^{\mathrm{L}}) T_{oc}(\psi_i^{\mathrm{L}}) T_{of}(\psi_i^{\mathrm{L}}) \cos(\psi_i^{\mathrm{L}}), \quad \forall i \in \mathcal{L}, \tag{8}$$

where m, A_{re} and $d_{i,u}$ denote the Lambertian index, the area of the PD and the distance between the i-th LED and the receiver. m is related with the LED half-illuminance semi-angle $\Theta_{1/2}$ and the relationship can be characterized by $m = -1/\log_2(\cos(\Theta_{1/2}))$. ϕ_i^{L} and ψ_i^{L} are the irradiance angle and incidence angle for the i-th LED, respectively. $T_{oc}(\cdot)$ and $T_{of}(\cdot)$ means the optical concentrate gain and optical filter gain, respectively. $T_{of}(\cdot)$ is usually set as a constant whereas the optical concentrate gain $T_{oc}(\cdot)$ is related with the refractive index in the PD and the field-of-view (FoV). The relationship can be given by

$$T_{oc}(\psi_i^{\mathrm{L}}) = \begin{cases} \alpha^2/\sin^2(\Psi_{\mathrm{FoV}}), & 0 \le \psi_i^{\mathrm{L}} \le \Psi_{\mathrm{FoV}}, \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

where α is the refractive index and Ψ_{FoV} is the FoV. Consequently, the LoS channel vector can be expressed as $\mathbf{h}^{\mathrm{LoS}} \triangleq [h_1^{\mathrm{LoS}}, \ldots, h_{N_L}^{\mathrm{LoS}}]^\top$.

For the i-th NLoS channel gain, the RIS-constructed NLoS channel model can refer to [22]. An assignment matrix is introduced to characterize the channel matrix. Details are as follows.

According to the analysis in [22], the optical RIS-reflected channel is suitable to be modeled as a spherical wave model rather than a planar wave model. It is more likely to be a near-field propagation following an "additive" model instead of a far-field model obeying a "multiplicative" form. Meanwhile, considering the nanoscale wavelength and transmission characteristic of the lightwave, the optical RIS-reflected channel can be approximately regarded as following geometric optics [34]. The reflected signal exhibits a concentrated power density predominantly along the direction dictated by the laws of reflection, with negligible energy leakage occurring in any deviating directions. Based on the analysis, if a cascaded LED-RIS-PD channel is established for a specific LED and an RIS reflecting element by tuning the coefficients, no other LED can align with this reflecting element and the PD due to the law of reflection. Consequently, we can define an assignment matrix to determine the matching relation among LEDs, RIS elements and the receiver.

To this end, given an assignment matrix $\mathbf{G} = [\mathbf{g}_1, \mathbf{g}_2, \ldots, \mathbf{g}_i, \ldots \mathbf{g}_{N_L}] \in \{0, 1\}^{N_r \times N_L}$, the channel matrix can be further formulated. N_r is the number of the optical RIS elements and $g_{n,i} = 1$ means the transmission signal from the i-th LED is reflected by the n-th RIS unit. Meanwhile, according to the analysis in [35], when the size of the LED is significantly smaller than the propagation distance, the RIS-reflected energy will become concentrated within a narrow beam. Based on this, it is assumed that one RIS unit can be assigned at most to one LED, which results in the following constraint

$$\sum_{i=1}^{N_L} g_{n,i} \leq 1, \quad \forall i \in \mathcal{L} \tag{10a}$$

$$g_{n,i} \in \{0, 1\}, \quad \forall n \in \mathcal{R} \tag{10b}$$

where $\mathcal{R} \triangleq \{1, \ldots, N_r\}$ denotes the set of the RIS elements. Based on the additive model, the reflective channel gain from the i-th LED to the receiver via the n-th RIS element can be expressed as

$$h_{n,i}^{\mathrm{NLoS}} = \delta \frac{(m+1)A_{re}}{2\pi(d_{n,i}^{\mathrm{R}} + d_{n,u}^{\mathrm{R}})^2} \cos^m(\phi_i^{\mathrm{R}}) \cos(\psi_i^{\mathrm{R}}) T_{oc}(\psi_i^{\mathrm{R}}) T_{of}(\psi_i^{\mathrm{R}}), \tag{11}$$

where δ denotes the reflective coefficient of the RIS unit and assumed to be a constant in our works, $d_{n,i}^{\mathrm{R}}$ and $d_{n,u}^{\mathrm{R}}$ are the distance from the i-th LED to the n-th RIS element, and the n-th RIS element to the receiver, respectively. ϕ_i^{R} and ψ_i^{R} are the irradiance angle and incidence angle for the i-th LED-to-n-th RIS element-to receiver path, respectively. $T_{oc}(\cdot)$ and $T_{of}(\cdot)$ can refer to (8). Consequently, the NLoS channel gain for the i-th LED can be formulated as

$$h_i^{\mathrm{NLoS}} = (\mathbf{h}_i^{\mathrm{r}})^\top \mathbf{g}_i, \quad \forall i \in \mathcal{L}, \tag{12}$$

where $\mathbf{h}_i^{\mathrm{r}} = [h_{1,i}^{\mathrm{NLoS}}, \ldots, h_{N_r,i}^{\mathrm{NLoS}}]^\top \in \mathbb{R}_{N_r \times 1}^+$. Considering the NLoS channel matrix $\mathbf{h}^{\mathrm{NLoS}} \triangleq [h_1^{\mathrm{NLoS}}, \ldots, h_{N_L}^{\mathrm{NLoS}}]^\top \in \mathbb{R}_{N_L \times 1}^+$, let $\mathbf{H} = [\mathbf{h}_1^{\mathrm{r}}, \ldots, \mathbf{h}_{N_L}^{\mathrm{r}}] \in \mathbb{R}_{N_r \times N_L}^+$, the NLoS channel vector can be expressed as

$$\mathbf{h}^{\mathrm{NLoS}} = \mathrm{diag}(\mathbf{H})^\top \mathrm{vec}(\mathbf{G}). \tag{13}$$

Consequently, let $\mathbf{I}^{\mathrm{LoS}} \triangleq [\mathbb{I}_1^{\mathrm{LoS}}, \ldots, \mathbb{I}_{N_L}^{\mathrm{LoS}}]^\top$, $\mathbf{I}^{\mathrm{NLoS}} \triangleq [\mathbb{I}_1^{\mathrm{NLoS}}, \ldots, \mathbb{I}_{N_L}^{\mathrm{NLoS}}]^\top$, the channel matrix of the VLCP system with multi-LED can be characterized by

$$\mathbf{h} = \mathbf{I}^{\mathrm{LoS}} \odot \mathbf{h}^{\mathrm{LoS}} + \mathbf{I}^{\mathrm{NLoS}} \odot \mathbf{h}^{\mathrm{NLoS}}. \tag{14}$$

The channel model is constructed until now.

2.2. Positioning and measurement

This part is divided into two parts: LoS positioning and NLoS positioning. The former is based on the fact that the LoS links are present and RIS is offline correspondingly, whereas the latter is due to there are blockages and the RIS is employed to provide added NLoS links to transmit the positioning information. The receiver accepts the positioning signals during the positioning process and estimate the RSS information according the received positioning signals. Details are as follows:

2.2.1. LoS positioning

For this part, the positioning information is acquired based on the LoS transmission in VLP. The RIS is offline in this scenario and the NLoS channel gain is zero, based on the Lambertian formula, the location information can be estimated based on (2) and (8).

According to (2), the electrical power of the i-th received positioning signal can be expressed as

$$\mathbb{E}\left\{y_{p,i}^2(t)\right\} = p_{p,i}(h_i^{\mathrm{LoS}})^2 + B_p N_0, \quad \forall i \in \mathcal{L}. \tag{15}$$

Since the channel gain is a function of the position of receiver, the position information of the receiver can be derived according to (8). The LoS channel gain can be reformulated as

$$h_i^{\mathrm{LoS}} = \frac{\beta}{||\mathbf{l}_i - \mathbf{u}||_2^2}\left(\frac{(\mathbf{u} - \mathbf{l}_i)^\top \mathbf{n}_i}{||\mathbf{l}_i - \mathbf{u}||_2}\right)^m \frac{(\mathbf{l}_i - \mathbf{u})^\top \mathbf{n}_u}{||\mathbf{l}_i - \mathbf{u}||_2}, \quad \forall i \in \mathcal{L}, \tag{16}$$

where $\beta = \frac{(m+1)A_{re}T_{oc}(\psi_i^{\mathrm{L}})T_{of}(\psi_i^{\mathrm{L}})}{2\pi}$, $\mathbf{l}_i$ and $\mathbf{u}$ are the locations of the i-th LED and the receiver, $\mathbf{n}_i$ and $\mathbf{n}_u$ are the orientations of the i-th LED and the receiver, respectively.

Assuming the orientations of LEDs are downwards and the receiver is upwards (i.e., $\mathbf{n}_i = (0, 0, -1)$ and $\mathbf{n}_u = (0, 0, 1)$), the channel gain can be expressed as

$$h_i^{\mathrm{LoS}} = \frac{\beta(z_{l_i} - z_u)^{m+1}}{||\mathbf{l}_i - \mathbf{u}||_2^{m+3}} = \sqrt{\frac{\mathbb{E}\left\{y_{p,i}^2(t)\right\} - B_p N_0}{p_{p,i}}}, \quad \forall i \in \mathcal{L}, \tag{17}$$

where z_{l_i} and z_u are the z-axis coordinates of the i-th LED and the receiver. Combined with (15) and (17), the location of the receiver can be estimated based on the received positioning signals and the locations of LEDs. The estimated position information can be applied for the channel estimation in the next period.

2.2.2. RIS positioning

When the LoS path is blocked, the RIS can provide extra NLoS links and transmit the positioning signals from LEDs, thereby preventing the interrupting of the positioning process. In this part, assuming that the LoS link for the i-th LED is blocked and the RIS provide an added NLoS link (i.e., $h_i = h_i^{\mathrm{NLoS}}$), the positioning information of the receiver can be estimated based on the relationship between the channel gain and the received positioning signals, similarly.

In this case, the received power of the i-th positioning signal is characterized by

$$\mathbb{E}\left\{y_{p,i}^2(t)\right\} = p_{p,i}(h_i^{\mathrm{NLoS}})^2 + B_p N_0, \quad \forall i \in \mathcal{L}. \tag{18}$$

Since the channel gain is a function of the position of the receiver, the channel gain can be derived according to the received positioning signals and the position can be solved accordingly. The derivation is as follows. Firstly, the reflective channel gain from the i-th LED to the receiver via the n-th RIS element in (11) can be rewritten as

$$h_{n,i}^{\mathrm{NLoS}} \frac{\zeta}{(||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2 + ||\tilde{\mathbf{l}}_n - \mathbf{u}||_2)^2}\left(\frac{(\tilde{\mathbf{l}}_n - \mathbf{l}_i)^\top \mathbf{n}_i}{||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2}\right)^m \frac{(\tilde{\mathbf{l}}_n - \mathbf{u})^\top \mathbf{n}_u}{||\tilde{\mathbf{l}}_n - \mathbf{u}||_2}, \forall i \in \mathcal{L}, \forall n \in \mathcal{R}, \tag{19}$$

where $\zeta = \frac{\delta(m+1)A_{re}T_{oc}(\psi_i^{\mathrm{R}})T_{of}(\psi_i^{\mathrm{R}})}{2\pi}$, $\tilde{\mathbf{l}}_n$ is the location of the n-th RIS element.

Similarly, the orientations of LEDs are assumed to be downwards and the receiver is upwards. Consequently, the formula can be expressed as

$$\begin{aligned} h_{n,i}^{\mathrm{NLoS}} &= \frac{\zeta(z_{l_i} - \tilde{z}_n)^m(\tilde{z}_n - z_u)}{(||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2 + ||\tilde{\mathbf{l}}_n - \mathbf{u}||_2)^2 ||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2^m ||\tilde{\mathbf{l}}_n - \mathbf{u}||_2} \\ &= \frac{\zeta(z_{l_i} - \tilde{z}_n)^m(\tilde{z}_n - z_u)}{||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2^{m+2}||\tilde{\mathbf{l}}_n - \mathbf{u}||_2 + ||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2^m||\tilde{\mathbf{l}}_n - \mathbf{u}||_2^3 + 2||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2^{m+1}||\tilde{\mathbf{l}}_n - \mathbf{u}||_2^2}, \end{aligned} \tag{20}$$

where $\tilde{z}_n$ is the z-axis coordinate of the n-th RIS element. Due to the fact that the locations of the LED and RIS element are known, the positioning information of the receiver can be estimated according to the received RIS-reflected positioning signals when the LoS paths are blocked.

Combined with (12), (18) and (20), the location can be estimated for the i-th NLoS positioning signal when the assignment matrix is determined.

Combined with the equations above, the position of the receiver can be estimated from the received signal power in the receiver. The relationship can be expressed as

$$h_i^{\mathrm{NLoS}} = \sum_{n=1}^{N_r} h_{n,i}^{\mathrm{NLoS}} g_{n,i} = \sqrt{\frac{\mathbb{E}\left\{y_{p,i}^2(t)\right\} - B_p N_0}{p_{p,i}}}, \quad \forall i \in \mathcal{L}. \tag{21}$$

When the location of the receiver is completely unknown (this is uncommon since there are other LoS paths even if the i-th LoS path is blocked, however, there is such a possibility), the assignment matrix is chosen randomly to obtain a rough position of the receiver. However, if the positioning information is partially known, the assignment matrix can be optimized to improve the location accuracy. In our works, we'd like to increase the received power to avoid the complex optimization for the CRLB.

Assuming the positioning error follows the Gaussian distribution [36], the error can be expressed as

$$\mathbf{e_u} = \mathbf{u} - \hat{\mathbf{u}}, \tag{22}$$

where $\hat{\mathbf{u}} = [\hat{u}_x, \hat{u}_y, \hat{u}_z]$ denotes the estimated position of the receiver and $\mathbf{e_u} = [e_x, e_y, e_z]$ denotes the positioning error subject to Gaussian distribution with mean $\mathbf{0}$ and covariance matrix $\mathbf{R}_p$. The variance of the position should be satisfied with the constraint on CRLB, which will be derived in Section 3.1.

2.3. Channel estimation

In the period of uplink transmission, the positioning information of the receiver is transmitted to the LEDs and the channel state information can be obtained according to the estimated position of the receiver. However, since the existence of the positioning error, the CSI is imperfect. Let $\hat{\mathbf{h}} = [\hat{h}_1, \hat{h}_2, \ldots, \hat{h}_i, \ldots, \hat{h}_{N_L}]^\top$ denotes the estimated CSI, and then, the CSI error can be given by $\mathbf{e_h} = \mathbf{h} - \hat{\mathbf{h}}$. Specifically, the i-th estimated channel gain can be expressed as

$$\hat{h}_i = \mathbb{I}_i^{\mathrm{LoS}} h_i^{\mathrm{LoS}}(\hat{\mathbf{u}}) + \mathbb{I}_i^{\mathrm{NLoS}} h_i^{\mathrm{NLoS}}(\hat{\mathbf{u}}), \quad \forall i \in \mathcal{L}, \tag{23}$$

and the CSI error of the i-th channel can be characterized by

$$\begin{aligned} e_{h_i} &= h_i - \hat{h}_i \\ &= \mathbb{I}_i^{\mathrm{LoS}} h_i^{\mathrm{LoS}}(\hat{\mathbf{u}} + \mathbf{e}_u) + \mathbb{I}_i^{\mathrm{NLoS}} h_i^{\mathrm{NLoS}}(\hat{\mathbf{u}} + \mathbf{e}_u) \\ &\quad - \mathbb{I}_i^{\mathrm{LoS}} h_i^{\mathrm{LoS}}(\hat{\mathbf{u}}) - \mathbb{I}_i^{\mathrm{NLoS}} h_i^{\mathrm{NLoS}}(\hat{\mathbf{u}}), \quad \forall i \in \mathcal{L}, \end{aligned} \tag{24}$$

where $\mathbf{e_h} = (e_{h_1}, \ldots, e_{h_{N_L}})$. From the equation, we can observed that the CSI error is a function of the position error.

According to (24), if we define that $e_{h_i} = q_i(\mathbf{e_u})$, and then, the probability density function of the CSI error of the i-th channel can be expressed as

$$f_{e_{h_i}}(e_{h_i}) = f_{\mathbf{e_u}}(q_i^{-1}(e_{h_i}))|\frac{\partial q_i^{-1}(e_{h_i})}{\partial e_{h_i}}|, \quad \forall i \in \mathcal{L}, \tag{25}$$

where $q_i^{-1}(e_{h_i})$ is the inverse function of (24). Consequently, the communication performance metric can be calculated according to the probability density function of the CSI error and the data transmission process will be influenced correspondingly.

Optics EXPRESS

2.4. Data transmission process

In the period of communication, the data is transmitted via the same frequency for different LEDs based on the CSI estimated above. We'd like to choose pulse amplitude modulation (PAM) as the modulate mode in the communication process for the advantage of high spectral efficiency and simplicity in practical.

The achievable data rate with PAM is considered in our work to represent the communication performance which can be characterized by [37]

$$R_c = B_c \log_2(1 + \frac{exp(1)}{2\pi}\frac{\kappa|(\hat{\mathbf{h}} + \mathbf{e_h})^\top \mathbf{w}_c|^2}{qB_cN_0}), \tag{26}$$

where q is the ratio converting transmitted optical power to electrical power, and κ is the PD responsivity coefficient set as a constant usually. Consequently, the achievable rate is a function of the signal power of the communication power division coefficient vector $\mathbf{w}_c$, the estimated channel gain $\hat{\mathbf{h}}$ and the CSI error $\mathbf{e_h}$.

Note that the estimated channel gain is obtained based on the estimated position of the receiver, which is acquired from the received positioning signal. Meanwhile, the received positioning signal is determined by the positioning power and RIS assignment matrix. Consequently, the estimated channel gain is a function of the power of the positioning signals and RIS assignment martix(i.e., determined by the positioning power division coefficient vector $\mathbf{w}_p = [\sqrt{p_{p,1}}, \sqrt{p_{p,2}}, \dots, \sqrt{p_{p,i}}, \dots, \sqrt{p_{p,N_L}}]^\top$ and $\mathbf{G}$). The CSI error is also determined by the error of the estimated receiver's position. Consequently, the achievable data rate is influenced by both the positioning power division coefficient $\mathbf{w}_p$ and the communication power division coefficient $\mathbf{w}_c$. Due to the constrained total LED power, the phenomenon ignites us to design the power allocation carefully in the VLCP system to achieve better system performance.

The position error is assumed to be satisfied with Gaussian distribution (i.e., $\mathbf{e_u} \sim \mathcal{N}(0, \mathbf{R}_p)$), the covariance matrix will be satisfied with the CRLB constraint. This constraint is correlated with the positioning power and RIS assignment martix, which will be discussed in Section 3.1.

3. Performance metrics

3.1. Cramer-Rao lower bound

We'd like to derive the CRLB of the positioning scheme we propose above in this part. The CRLB is discussed in this paper to quantify the accuracy of the proposed positioning scheme. It is observed that the variance of the position error should be satisfied with [38]

$$\mathbb{E}\left\{||\mathbf{e_u}||^2\right\} = \mathrm{Tr}(\mathbf{R}_p) \geq \mathrm{Tr}(\mathbf{J_u}^{-1}(\mathbf{w}_p, \mathbf{G})), \tag{27}$$

where $\mathbf{J_u}(\mathbf{w}_p, \mathbf{G})$ is the Fisher Information Matrix (FIM), which can be calculated by the log-likelihood function of the received signals. The derivation process is as follows.

The likelihood function of the received positioning signal from the i-th LED is expressed as

$$f(y_{p,i}(t); \mathbf{u}) = \frac{1}{\sqrt{2\pi B_p N_0}} e^{-\frac{\int_0^{T_p}(y_{p,i}(t) - h_i\sqrt{p_{p,i}}s_p(t)\cos(2\pi f_{p,i}t))^2 \mathrm{d}t}{2B_pN_0}}. \tag{28}$$

Consequently, the log-likelihood function can be given by

$$\begin{aligned}\Upsilon(\mathbf{u}) &= \log(\prod_{i=1}^{N_L} f(y_{p,i}(t); \mathbf{u})) \\ &= \log\iota - \frac{1}{2B_pN_0}\sum_{i=1}^{N_L}\int_0^{T_p}(y_{p,i}(t) \\ &- h_i\sqrt{p_{p,i}}s_p(t)\cos(2\pi f_{p,i}t))^2\mathrm{d}t,\end{aligned} \tag{29}$$

where ι denotes the parameters which is irrelevant with $\mathbf{u}$. The estimated receiver's location $\hat{\mathbf{u}}$ is characterized as

$$\hat{\mathbf{u}} = \arg\max_{\mathbf{u}} \Upsilon(\mathbf{u}). \tag{30}$$

The FIM for the estimated position can be defined as [39]

$$[\mathbf{J_u}]_{k,l} = -\mathbb{E}\left\{\frac{\partial^2 \Upsilon(\mathbf{u})}{\partial \mathbf{u}_k \partial \mathbf{u}_l}\right\}, \tag{31}$$

where $k, l \in \{1, 2, 3\}$ and $\mathbf{u}_k$ denotes the coordinate of the receiver in the k-th coordinate axis. The FIM can be further expressed as

$$\mathbf{J_u} = -\begin{bmatrix} \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial x_u^2}\right\} & \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial x_u \partial y_u}\right\} & \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial x_u \partial z_u}\right\} \\ \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial y_u \partial x_u}\right\} & \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial y_u^2}\right\} & \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial y_u \partial z_u}\right\} \\ \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial z_u \partial x_u}\right\} & \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial z_u \partial y_u}\right\} & \mathbb{E}\left\{\frac{\partial^2\Upsilon(\mathbf{u})}{\partial z_u^2}\right\} \end{bmatrix} \tag{32}$$

where x_u and y_u denote the x-axis and y-axis coordinates of the receiver, respectively. The first partial derivatives of the log-likelihood function can be expressed as

$$\begin{aligned} \frac{\partial \Upsilon(\mathbf{u})}{\partial \mathbf{u}_k} = &-\frac{1}{B_p N_0}\sum_{i=1}^{N_L}\int_0^{T_p}(h_i p_{p,i} s_p^2(t)\cos^2(2\pi f_{p,i} t) \\ &- y_{p,i}(t)\sqrt{p_{p,i}} s_p(t)\cos(2\pi f_{p,i} t))\frac{\partial h_i}{\partial \mathbf{u}_k}\mathrm{d}t, \end{aligned} \tag{33}$$

where $k \in \{1, 2, 3\}$. Then, the second partial derivatives of the log-likelihood function can be expressed as

$$\begin{aligned} \frac{\partial^2 \Upsilon(\mathbf{u})}{\partial \mathbf{u}_k^2} = &-\frac{1}{B_p N_0}\sum_{i=1}^{N_L}\int_0^{T_p}((\frac{\partial h_i}{\partial \mathbf{u}_k})^2 p_{p,i} s_p^2(t)\cos^2(2\pi f_{p,i} t) \\ &+ h_i p_{p,i} s_p^2(t)\cos^2(2\pi f_{p,i} t)\frac{\partial^2 h_i}{\partial \mathbf{u}_k^2} \\ &- y_{p,i}(t)\sqrt{p_{p,i}} s_p(t)\cos(2\pi f_{p,i} t)\frac{\partial^2 h_i}{\partial \mathbf{u}_k^2})\mathrm{d}t, \end{aligned} \tag{34}$$

$$\begin{aligned} \frac{\partial^2 \Upsilon(\mathbf{u})}{\partial \mathbf{u}_k \partial \mathbf{u}_l} = &-\frac{1}{B_p N_0}\sum_{i=1}^{N_L}\int_0^{T_p}(\frac{\partial h_i}{\partial \mathbf{u}_k}\frac{\partial h_i}{\partial \mathbf{u}_l} p_{p,i} s_p^2(t)\cos^2(2\pi f_{p,i} t) \\ &+ h_i p_{p,i} s_p^2(t)\cos^2(2\pi f_{p,i} t)\frac{\partial h_i}{\partial \mathbf{u}_k}\frac{\partial h_i}{\partial \mathbf{u}_l} \\ &- y_{p,i}(t)\sqrt{p_{p,i}} s_p(t)\cos(2\pi f_{p,i} t)\frac{\partial h_i}{\partial \mathbf{u}_k}\frac{\partial h_i}{\partial \mathbf{u}_l})\mathrm{d}t. \end{aligned} \tag{35}$$

Since the constraints on the positioning signal, we can know that $\mathbb{E}\{s_p(t)\} = 0$ and $\mathbb{E}\{s_p^2(t)\} = 1$. Consequently, the term in FIM can be expressed as

$$\begin{aligned} \mathbb{E}\left\{\frac{\partial^2 \Upsilon(\mathbf{u})}{\partial \mathbf{u}_k^2}\right\} &= -\frac{T_p}{2B_p N_0}\sum_{i=1}^{N_L} p_{p,i}(\frac{\partial h_i}{\partial \mathbf{u}_k})^2, \\ \mathbb{E}\left\{\frac{\partial^2 \Upsilon(\mathbf{u})}{\partial \mathbf{u}_k \partial \mathbf{u}_l}\right\} &= -\frac{T_p}{2B_p N_0}\sum_{i=1}^{N_L} p_{p,i}\frac{\partial h_i}{\partial \mathbf{u}_k}\frac{\partial h_i}{\partial \mathbf{u}_l}. \end{aligned} \tag{36}$$

According to (5), we can obtain that

$$\frac{\partial h_i}{\partial \mathbf{u}_k} = \mathbb{I}_i^{\mathrm{LoS}} \frac{\partial h_i^{\mathrm{LoS}}}{\partial \mathbf{u}_k} + \mathbb{I}_i^{\mathrm{NLoS}} \frac{\partial h_i^{\mathrm{NLoS}}}{\partial \mathbf{u}_k}. \tag{37}$$

According to (17) and (20), it is observed that

$$\begin{aligned}
\frac{\partial h_i^{\mathrm{LoS}}}{\partial x_u} &= \frac{-\beta(m+3)(z_{l_i}-z_u)^{m+1}(x_u-x_{l_i})}{||\mathbf{l}_i-\mathbf{u}||_2^{m+5}},\\
\frac{\partial h_i^{\mathrm{LoS}}}{\partial y_u} &= \frac{-\beta(m+3)(z_{l_i}-z_u)^{m+1}(y_u-y_{l_i})}{||\mathbf{l}_i-\mathbf{u}||_2^{m+5}},\\
\frac{\partial h_i^{\mathrm{LoS}}}{\partial z_u} &= \frac{-\beta(m+1)(z_{l_i}-z_u)^{m}}{||\mathbf{l}_i-\mathbf{u}||_2^{m+3}} + \frac{\beta(m+3)(z_{l_i}-z_u)^{m+2}}{||\mathbf{l}_i-\mathbf{u}||_2^{m+5}},
\end{aligned} \tag{38}$$

$$\begin{aligned}
\frac{\partial h_{n,i}^{\mathrm{NLoS}}}{\partial x_u} &= \frac{\zeta(z_{l_i}-\tilde{z}_n)^m(\tilde{z}_n-z_u)(\tilde{x}_n-x_u)}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^4||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m-1}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2^2}\left\{\frac{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}+3\frac{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}+4\right\}\\
\frac{\partial h_{n,i}^{\mathrm{NLoS}}}{\partial y_u} &= \frac{\zeta(z_{l_i}-\tilde{z}_n)^m(\tilde{z}_n-z_u)(\tilde{y}_n-y_u)}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^4||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m-1}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2^2}\left\{\frac{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}+3\frac{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}+4\right\}\\
\frac{\partial h_{n,i}^{\mathrm{NLoS}}}{\partial z_u} &= \frac{\zeta(z_{l_i}-\tilde{z}_n)^m(\tilde{z}_n-z_u)^2}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^4||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m-1}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2^2}\left\{\frac{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}+3\frac{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}+4\right\}\\
&\quad - \frac{\zeta(z_{l_i}-\tilde{z}_n)^m}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^2||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}.
\end{aligned} \tag{39}$$

According to (12), it is observed that

$$\frac{\partial h_i^{\mathrm{NLoS}}}{\partial \mathbf{u}_k} = [\frac{\partial \mathbf{h}_i^{\mathrm{r}}}{\partial \mathbf{u}_k}]^\top \mathbf{g}_i, \quad \forall i \in \mathcal{L}, \tag{40}$$

where

$$\frac{\partial \mathbf{h}_i^{\mathrm{r}}}{\partial \mathbf{u}_k} = [\frac{\partial h_{1,i}^{\mathrm{NLoS}}}{\partial \mathbf{u}_k}, \dots, \frac{\partial h_{N_r,i}^{\mathrm{NLoS}}}{\partial \mathbf{u}_k}]^\top \in \mathbb{R}_{N_r\times 1}^+.$$

Considering the NLoS partial channel matrix

$$\frac{\partial \mathbf{h}^{\mathrm{NLoS}}}{\partial \mathbf{u}_k} \triangleq [\frac{\partial h_1^{\mathrm{NLoS}}}{\partial \mathbf{u}_k}, \dots, \frac{\partial h_{N_L}^{\mathrm{NLoS}}}{\partial \mathbf{u}_k}]^\top \in \mathbb{R}_{N_L\times 1}^+,$$

let $\frac{\partial \mathbf{H}}{\partial \mathbf{u}_k} = [\frac{\partial \mathbf{h}_1^{\mathrm{r}}}{\partial \mathbf{u}_k}, \dots, \frac{\partial \mathbf{h}_{N_L}^{\mathrm{r}}}{\partial \mathbf{u}_k}] \in \mathbb{R}_{N_r\times N_L}^+$, the NLoS partial channel vector can be expressed as

$$\frac{\partial \mathbf{h}^{\mathrm{NLoS}}}{\partial \mathbf{u}_k} = \mathrm{diag}(\frac{\partial \mathbf{H}}{\partial \mathbf{u}_k})^\top \mathrm{vec}(\mathbf{G}). \tag{41}$$

According to (38) and (40), the quantification of (37) can be calculated, and meanwhile, the FIM and CRLB constraint can be derived according to the equations above. The CRLB is considered as the positioning performance indicator to evaluate the proposed VLCP system.

3.2. Outage probability

In our works, the outage probability is considered as the communication performance indicator to evaluate the proposed VLCP system. According to (26), we can obtain that the achievable rate is a function of the CSI error $\mathbf{e_h}$, and meanwhile, the CSI error is a function of the positioning error affected by the positioning power. Consequently, the achievable rate is determined by both communication power and positioning power (i.e., $\mathbf{w}_p$ and $\mathbf{w}_c$).

Meanwhile, the achievable data rate is a function of RIS parameters, and consequently, the corresponding parameters which can influence the achievable data rate should be carefully adjusted to satisfy the constraints on the communication performance metric. Since the achievable data rate is a function of the positioning error, the achievable rate is satisfied with one probability distribution which can be calculated by the relationship between the achievable data rate and the positioning error.

What's more, the system is in outage if the achievable rate is less than a threshold. The probability of the achievable rate less than the threshold is regarded as the outage probability. The probability can be expressed as

$$\Pr\left\{R_c(\mathbf{w}_p, \mathbf{w}_c, \mathbf{G}) \leq R_0\right\} \leq P_{out}, \tag{42}$$

where R_0 represents the constraint on the quality of service (QoS) which denotes the threshold of the achievable rate, P_{out} denotes the maximum tolerate outage probability considered as the communication performance metric in our works.

3.3. Trade-off between communication and positioning

From the derivation above, we can obtain that the achievable rate is influenced by the power of both communication and positioning. The more power allocated to the positioning, the higher accuracy of VLP will be. However, this allocation reduces the power available for communication, potentially deteriorating the communication metric. Consequently, the power of the LED should be allocated carefully to make a trade-off between communication and positioning.

4. Optimization of VLCP system

In this section, we'd like to minimize the CRLB when the VLCP system is subject to the QoS constraint (i.e., the achievable rate is subject to a constraint guaranteeing the communication rate for the user cannot be in outage). Meanwhile, the power coefficient of communication and positioning, the assignment matrix of RIS, are the optimization parameters to minimize the optimization goal. The optimization problem can be characterized by

$$\mathbf{P:}\ \min_{\mathbf{w}_p, \mathbf{w}_c, \mathbf{G}} \mathrm{Tr}(\mathbf{J_u}^{-1}(\mathbf{w}_p, \mathbf{G})) \tag{43a}$$

$$s.t. \quad \Pr\left\{R_c(\mathbf{w}_p, \mathbf{w}_c, \mathbf{G}) \leq R_0\right\} \leq P_{out}, \tag{43b}$$

$$0 \leq \mathbf{e}_i^\top \mathbf{w}_p + \mathbf{e}_i^\top \mathbf{w}_c \leq \widetilde{P}, \tag{43c}$$

$$0 \leq \mathbf{e}_i^\top \mathbf{w}_p \mathbf{w}_p^\top \mathbf{e}_i + \mathbf{e}_i^\top \mathbf{w}_c \mathbf{w}_c^\top \mathbf{e}_i \leq P_e - I_{DC}^2, \tag{43d}$$

$$(10a), (10b) \tag{43e}$$

where $\mathbf{e}_i$ is a column vector with zeros except for the i-th element, $\widetilde{P} \triangleq \min\{\frac{I_{DC}}{A}, \frac{P_o - I_{DC}}{A}\}$ denotes the non-negative and peak-power constraint.

The adjustment of the LED power allocation and the RIS assignment matrix will change the probability distribution of the positioning error, the CSI error, and therefore, the achievable rate. Consequently, we can obtain the optimal CRLB given the outage probability constraint P_{out} through the optimization of the LED power allocation and the RIS assignment matrix. Since the optimization problem is intractable and non-convex, in this paper, we propose an alternative optimization (AO) algorithm to solve this problem.

4.1. Power coefficient optimization subproblem given the RIS assignment matrix

Given the fixed RIS assignment matrix (assuming that it is in the t-th iteration and the assignment matrix is labeled as $\mathbf{G}^{(t)}$), we can focus on the communication and positioning power coefficient optimization to minimize the CRLB under the outage probability constraint. The primary optimization problem can be simplified as

$$\min_{\mathbf{w}_p,\mathbf{w}_c} \mathrm{Tr}(\mathbf{J_u}^{-1}(\mathbf{w}_p, \mathbf{G}^{(t)}) \tag{44a}$$

$$s.t. \quad \Pr\left\{R_c(\mathbf{w}_p, \mathbf{w}_c, \mathbf{G}^{(t)} \leq R_0\right\} \leq P_{out}, \tag{44b}$$

$$0 \leq \mathbf{e}_i^\top \mathbf{w}_p + \mathbf{e}_i^\top \mathbf{w}_c \leq \widetilde{P}, \tag{44c}$$

$$0 \leq \mathbf{e}_i^\top \mathbf{w}_p \mathbf{w}_p^\top \mathbf{e}_i + \mathbf{e}_i^\top \mathbf{w}_c \mathbf{w}_c^\top \mathbf{e}_i \leq P_e - I_{DC}^2. \tag{44d}$$

Before introducing the optimization process, we'd like to transform the non-convex constraints into tractable forms.

According to (26), the QoS constraint in (44b) can be expressed as

$$R_c \leq R_0 \iff |(\hat{\mathbf{h}} + \mathbf{e_h})^\top \mathbf{w}_c|^2 \leq \varsigma, \tag{45}$$

where $\varsigma = (2^{\frac{R_0}{B_c}-1})2\pi q B_c N_0/(\kappa exp(1))$. Assuming that $\mathbf{W}_c = \mathbf{w}_c\mathbf{w}_c^\top$, we can obtain $\mathbf{W}_c \succeq \mathbf{0}, \mathrm{rank}(\mathbf{W}_c) = 1$. Consequently, (45) can be reformulated as

$$\begin{aligned} \hat{\mathbf{h}}^\top \mathbf{W}_c \hat{\mathbf{h}} + 2\hat{\mathbf{h}}^\top \mathbf{W}_c \mathbf{e_h} + \mathbf{e_h}^\top \mathbf{W}_c \mathbf{e_h} \leq \varsigma, \\ \mathbf{W}_c \succeq \mathbf{0}, \mathrm{rank}(\mathbf{W}_c) = 1. \end{aligned} \tag{46}$$

Consequently, the constraint in (44b) can be rewritten as

$$\Pr\left\{\hat{\mathbf{h}}^\top \mathbf{W}_c \hat{\mathbf{h}} + 2\hat{\mathbf{h}}^\top \mathbf{W}_c \mathbf{e_h} + \mathbf{e_h}^\top \mathbf{W}_c \mathbf{e_h} \leq \varsigma\right\} \leq P_{out}, \tag{47a}$$

$$\mathbf{W}_c \succeq \mathbf{0}, \mathrm{rank}(\mathbf{W}_c) = 1. \tag{47b}$$

Note that the probability constraint in (47a) is complex, a good convex approximation named Conditional Value-at-Risk (CVaR) based method is introduced to deal with the probability constraint [40,41]. From (25), the explicitly probability distribution of CSI error is hard to calculate according to the distribution of the positioning error. However, we can calculate the expectation and variance of the CSI error based on the relationship between the CSI error and the positioning error. According to (25), the expectation of the CSI error for the i-th channel can be calculated by

$$\mu_i = \mathbb{E}\left\{e_{h_i}\right\} = \int q_i(\mathbf{e_u}) f_{\mathbf{e_u}}(\mathbf{e_u}) \mathrm{d}\mathbf{e_u}, \quad \forall i \in \mathcal{L}. \tag{48}$$

We choose $\boldsymbol{\mu} = [\mu_1, \ldots, \mu_{N_L}]^\top$ to denote the expectation of CSI error vector. Similarly, the covariance matrix of the CSI error can be calculated by

$$\boldsymbol{\Sigma} = \mathbb{E}\left\{(\mathbf{e_h} - \boldsymbol{\mu})(\mathbf{e_h} - \boldsymbol{\mu})^\top\right\}. \tag{49}$$

According to the CVaR method, the constraint in (47a) can be written as

$$\lambda + \frac{1}{P_{\text{out}}} \mathrm{Tr}(\boldsymbol{\Omega}\mathbf{D}) \leq 0 \tag{50a}$$

$$\mathbf{D} - \begin{bmatrix} -\mathbf{W}_c & -\mathbf{W}_c^\mathrm{T}\hat{\mathbf{h}} \\ -\hat{\mathbf{h}}^\mathrm{T}\mathbf{W}_c & -\hat{\mathbf{h}}^\mathrm{T}\mathbf{W}_c\hat{\mathbf{h}} + \varsigma - \lambda \end{bmatrix} \succeq \mathbf{0} \tag{50b}$$

Optics EXPRESS

$$\mathbf{D} \succeq \mathbf{0}, \mathbf{D} \in \mathbb{S}^4 \tag{50c}$$

where $\mathbf{D}$ and λ are the auxiliary variables. $\mathbb{S}^4$ denotes the 4th-order real symmetric matrix, and

$$\mathbf{\Omega} = \begin{bmatrix} \Sigma + \boldsymbol{\mu}\boldsymbol{\mu}^{\mathrm{T}} & \boldsymbol{\mu} \\ \boldsymbol{\mu}^{\mathrm{T}} & \mathbf{1} \end{bmatrix} \tag{51}$$

Consequently, by the CVaR method-based transform and ignoring the rank-1 constraint $\mathrm{rank}(\mathbf{W}_c) = 1$, the semidefinite relaxation (SDR) [42] can be explored to convert the constraint in (44b) into several convex constraints. Please refer to Appendix 7.1 for the proof of the transform.

Consequently, ignoring the rank-1 constraint, the power coefficient optimization subproblem given the RIS assignment matrix can be transformed as a semidefinite program (SDP) problem expressed as

$$\min_{\mathbf{w}_p, \mathbf{W}_c} \mathrm{Tr}(\mathbf{J_u}^{-1}(\mathbf{w}_p, \mathbf{G}^{(t)}) \tag{52a}$$

$$s.t. \quad 0 \le \mathbf{e}_i^{\top}\mathbf{w}_p + \sqrt{\mathrm{Tr}(\mathbf{W}_c\mathbf{e}_i\mathbf{e}_i^{\top})} \le \widetilde{P}, \tag{52b}$$

$$0 \le \mathbf{e}_i^{\top}\mathbf{w}_p\mathbf{w}_p^{\top}\mathbf{e}_i + \mathrm{Tr}(\mathbf{W}_c\mathbf{e}_i\mathbf{e}_i^{\top}) \le P_e - I_{DC}^2, \tag{52c}$$

$$\mathbf{W}_c \succeq \mathbf{0} \tag{52d}$$

$$(50a), (50b), (50c). \tag{52e}$$

Similarly, assuming that $\mathbf{W}_p = \mathbf{w}_p\mathbf{w}_p^{\top}$, we can know that $\mathbf{W}_p \succeq \mathbf{0}, \mathrm{rank}(\mathbf{W}_p) = 1$. By ignoring the rank-1 constraint, the optimization subproblem can be characterized by

$$\min_{\mathbf{W}_p, \mathbf{W}_c} \mathrm{Tr}(\mathbf{J_u}^{-1}(\mathbf{W}_p, \mathbf{G}^{(t)}) \tag{53a}$$

$$s.t. \quad 0 \le \sqrt{\mathrm{Tr}(\mathbf{W}_p\mathbf{e}_i\mathbf{e}_i^{\top})} + \sqrt{\mathrm{Tr}(\mathbf{W}_c\mathbf{e}_i\mathbf{e}_i^{\top})} \le \widetilde{P}, \tag{53b}$$

$$0 \le \mathrm{Tr}(\mathbf{W}_p\mathbf{e}_i\mathbf{e}_i^{\top}) + \mathrm{Tr}(\mathbf{W}_c\mathbf{e}_i\mathbf{e}_i^{\top}) \le P_e - I_{DC}^2, \tag{53c}$$

$$\mathbf{W}_c \succeq \mathbf{0} \tag{53d}$$

$$\mathbf{W}_p \succeq \mathbf{0} \tag{53e}$$

$$(50a), (50b), (50c). \tag{53f}$$

Note that the optimization object (53a) and the constraint in (53b) are non-convex. The optimization object can be transformed as a convex function by successive convex approximation (SCA). The first-order Taylor series expansion is applied to transform the constraint in (53b) into an affine approximation.

The optimization object can be conservatively transformed as

$$\max_{\mathbf{W}_p, \mathbf{W}_c} \mathrm{Tr}(\mathbf{J_u}(\mathbf{W}_p, \mathbf{G}^{(t)}). \tag{54}$$

According to Section 3.1, we can obtain that the $\mathrm{Tr}(\mathbf{J_u}(\mathbf{W}_p, \mathbf{G}^{(t)})$ is a linear function of $\mathbf{w}_p$. Consequently, the object function can be rewritten as

$$\max_{\mathbf{w}_p, \mathbf{W}_c} \mathbf{w}_p^{\top}\mathbf{c}, \tag{55}$$

where $\mathbf{c} = \sum_{k=1}^{3} \mathbf{c}_k$, and $\mathbf{c}_k = [c_{1,k}, \ldots, c_{N_L,k}]^\top \in \mathbb{R}^{+}_{N_L\times 1}$, $c_{i,k} = \frac{T_p}{2B_pN_0}(\frac{\partial h_i}{\partial \mathbf{u}_k})^2, k \in \{1,2,3\}, \forall i \in \mathcal{L}$. Let $\mathbf{C} = \mathrm{diag}(\mathbf{c} \odot \mathbf{c}) \in \mathbb{S}^{N_L}$, according to the Cauchy-Schwarz inequality, we can obtain that

$$(\mathbf{w}_p^\top \mathbf{c})^2 \leq 2\,\mathrm{Tr}(\mathbf{C}\mathbf{W}_p). \tag{56}$$

Consequently, the objective object can be conservatively transformed as

$$\min_{\mathbf{W}_p,\mathbf{W}_c} \mathrm{Tr}(\mathbf{C}\mathbf{W}_p). \tag{57}$$

Consequently, the optimization object can be conservatively transformed as a convex function versus $\mathbf{W}_p$. The constraint in (53b) can be approximately transformed as an affine constraint by the first-order Taylor series expansion characterized by

$$\begin{aligned}\sqrt{\mathrm{Tr}(\mathbf{W}_c\mathbf{e}_i\mathbf{e}_i^\top)} \approx & \sqrt{\mathrm{Tr}\left(\mathbf{W}_{c,0}\mathbf{e}_i\mathbf{e}_i^\top\right)} + \frac{1}{2}\left(\sqrt{\mathrm{Tr}\left(\mathbf{W}_{c,0}\mathbf{e}_i\mathbf{e}_i^\top\right)}\right)^{-1} \\ & \cdot \mathrm{Tr}\left(\left(\mathbf{W}_c - \mathbf{W}_{c,0}\right)\mathbf{e}_i\mathbf{e}_i^\top\right) \triangleq \tilde{w}_{c,i},\end{aligned} \tag{58}$$

$$\begin{aligned}\sqrt{\mathrm{Tr}(\mathbf{W}_p\mathbf{e}_i\mathbf{e}_i^\top)} \approx & \sqrt{\mathrm{Tr}\left(\mathbf{W}_{p,0}\mathbf{e}_i\mathbf{e}_i^\top\right)} + \frac{1}{2}\left(\sqrt{\mathrm{Tr}\left(\mathbf{W}_{p,0}\mathbf{e}_i\mathbf{e}_i^\top\right)}\right)^{-1} \\ & \cdot \mathrm{Tr}\left(\left(\mathbf{W}_p - \mathbf{W}_{p,0}\right)\mathbf{e}_i\mathbf{e}_i^\top\right) \triangleq \tilde{w}_{p,i},\end{aligned} \tag{59}$$

where $\mathbf{W}_{c,0}$ and $\mathbf{W}_{p,0}$ denote any given initial values. Consequently, the non-convex constraint can be approximately transformed as

$$0 \leq \tilde{w}_{c,i} + \tilde{w}_{p,i} \leq \tilde{P}. \tag{60}$$

The power coefficient optimization subproblem is transformed as a convex semidefinite program (SDP) expressed as

$$\mathbf{P1:} \min_{\mathbf{W}_p,\mathbf{W}_c} \mathrm{Tr}(\mathbf{C}\mathbf{W}_p) \tag{61a}$$

$$s.t. \quad (50a), (50b), (50c), (53c), (53d), (53e), (60). \tag{61b}$$

This problem is convex and can be calculated by the CVX toolbox in MATLAB [43]. If the solution is satisfied with the ignorant rank-1 constraints $\mathrm{rank}(\mathbf{W}_c) = 1$ and $\mathrm{rank}(\mathbf{W}_p) = 1$, the power coefficient vectors $\mathbf{w}_c$ and $\mathbf{w}_p$ can be solved by the eigenvalue decomposition of $\mathbf{W}_p$ and $\mathbf{W}_c$. If the constraint cannot be satisfied, the Gaussian randomization method is applied to find the satisfied solution [42].

4.2. RIS optimization subproblem given power coefficient

To avoid the complex optimization for the CRLB, the RIS parameters are optimized to enhance the received power. This is reasonable since that the enhancement of the received positioning power gives rise to the improvement of the positioning accuracy, and meanwhile, the promotion of the received communication power gives rise to the improvement of the achievable rate. Consequently, the RIS assignment matrix is optimized to maximize the sum of the received powers. Given the power coefficient, the RIS optimization subproblem can be expressed as

$$\max \mathbf{h}^\top(\mathbf{w}_p^{(t+1)} + \mathbf{w}_c^{(t+1)}) \tag{62a}$$

$$s.t. \quad \sum_{i=1}^{N_L} g_{n,i} \leq 1, \quad \forall i \in \mathcal{L} \tag{62b}$$

$$g_{n,i} \in \{0,1\}, \quad \forall n \in \mathcal{R} \tag{62c}$$

where $\mathbf{w}_p^{(t+1)}$ and $\mathbf{w}_c^{(t+1)}$ are the positioning and communication power coefficient at the $(t+1)$-th iteration and considered as the fixed constant in the optimization process of the RIS assignment matrix. Without loss of generality, we assume the $\mathbf{w}^{(t+1)} = \mathbf{w}_p^{(t+1)} + \mathbf{w}_c^{(t+1)}$, and consequently, the object function can be defined as

$$f(\mathbf{G}, \mathbf{w}^{(t+1)}) \triangleq \mathbf{h}^\top \mathbf{w}^{(t+1)}. \tag{63}$$

The optimization subproblem can be expressed as

$$\max f(\mathbf{G}, \mathbf{w}^{(t+1)}) \tag{64a}$$

$$s.t. \quad \sum_{i=1}^{N_L} g_{n,i} \leq 1, \quad \forall i \in \mathcal{L} \tag{64b}$$

$$g_{n,i} \in \{0, 1\}. \quad \forall n \in \mathcal{R} \tag{64c}$$

From (64a), we can observe that the objective function is linear with respect to $\mathbf{G}$. As a result of the IM scheme in VLC, the entries of $\mathbf{w}^{(t+1)}$ are real-valued and nonnegative, and therefore, $f(\mathbf{G}, \mathbf{w}^{(t+1)})$ increases monotonically with $g_{n,i}$. The constraint in (64b) can be rewritten as

$$\sum_{i=1}^{N_L} g_{n,i} = 1, \quad \forall i \in \mathcal{L}, \quad \forall n \in \mathcal{R}. \tag{65}$$

The optimization subproblem can be rewritten as

$$\textbf{P2:} \max f(\mathbf{G}, \mathbf{w}^{(t+1)}) \tag{66a}$$

$$s.t. \quad \sum_{i=1}^{N_L} g_{n,i} = 1, \quad \forall i \in \mathcal{L} \tag{66b}$$

$$g_{n,i} \geq 0, \quad \forall n \in \mathcal{R} \tag{66c}$$

From (13) and (14), the objective function can be analyzed as follows.

$$\begin{aligned} f(\mathbf{G}, \mathbf{w}^{(t+1)}) &= (\mathbf{w}^{(t+1)})^\top \mathbf{h}^{\mathrm{LoS}} + (\mathbf{w}^{(t+1)})^\top \mathrm{diag}(\mathbf{H})^\top \mathrm{vec}(\mathbf{G}) \\ &= \sum_{n=1}^{N_r} \sum_{i=1}^{N_L} h_{n,i}^{\mathrm{NLoS}} g_{n,i} w_i^{(t+1)} + (\mathbf{w}^{(t+1)})^\top \mathbf{h}^{\mathrm{LoS}} \\ &\leq \sum_{n=1}^{N_r} \max_i h_{n,i}^{\mathrm{NLoS}} w_i^{(t+1)} + (\mathbf{w}^{(t+1)})^\top \mathbf{h}^{\mathrm{LoS}}, \end{aligned}$$

where $w_i^{(t+1)}$ is the i-th item of $\mathbf{w}^{(t+1)}$, and the equality holds when the index of LED is chosen based on the principle achieving the maximization of the product $h_{n,i}^{\mathrm{NLoS}} w_i^{(t+1)}$. In other words, the index of LED is determined as

$$\tilde{i} = \arg\max_i h_{n,i}^{\mathrm{NLoS}} w_i^{(t+1)}, \quad \forall n \in \mathcal{R}. \tag{67}$$

Consequently, based on the kind of chosen principle, the n-th row of the chosen assignment matrix $\tilde{\mathbf{G}}$ is determined by

$$\tilde{g}_{n,i} = \begin{cases} 1, & \text{if } i = \tilde{i} \\ 0, & \text{otherwise} \end{cases} \tag{68}$$

It is observed that $\tilde{\mathbf{G}}$ is subject to the constraint in (10a) and (10b). Meanwhile, the chosen assignment matrix can achieve the maximization of the objective function of the subproblem. Consequently, we choose the maximum received power chosen (MRPC) principle to determine the RIS assignment matrix given the power coefficient.

4.3. Alternative optimization algorithm to proposed optimization problem

Based on the CVaR-based approximation, SCA and SDR methods, the power coefficient optimization subproblem given a fixed RIS assignment matrix is conservatively relaxed as a convex problem **P1** able to be solved by interior point method or CVX toolbox in polynomial time. Based on the result, the eigenvalue decomposition or Gaussian randomization is applied to complement the gap between the relaxed solution and the satisfied solution to the primary problem.

Meanwhile, the RIS assignment matrix optimization subproblem given fixed power coefficient **P2** is solved based on the MRPC to maximize the received power of the receiver so as to avoid the complex optimization process on CRLB. Combined with the two subproblems, we proposed an AO algorithm to solve the proposed optimization problem to minimize the CRLB with corresponding communication performance metric constraint, power constraint and RIS constraint. Algorithm 1 summarizes the overall optimization process of the proposed optimization problem. In the algorithm, the initialization of the RIS assignment matrix is based on the minimum distance criterion.

Algorithm 1. Proposed AO algorithm for CRLB minimization

Input: Iteration termination condition $\epsilon \geq 0$, outage probability P_{out}, threshold R_0, power constraint P_e, $\tilde{P}$, I_{DC}, iteration index $t \leftarrow 0$

Output: Optimal power coefficient $\mathbf{w}_p^*$, $\mathbf{w}_c^*$, and RIS assignment matrix $\mathbf{G}^*$

1: Initialize the assignment matrix at the t-th iteration $\mathbf{G}^{(t)}$ based on the minumum distance criterion
$\mathbf{G}^{(t)} : g_{n,i} \leftarrow 1, \text{if } i = \arg\min_i d^{\mathrm{R}}_{n,i};\ g_{n,j} \leftarrow 0, \forall j \neq i;$
Initialize the power condition-satisfied coefficient $\mathbf{W}_{c,0}$ and $\mathbf{W}_{p,0}$;
2: **repeat**
3: Calculate the transformed relaxed convex problem **P1**;
4: Solve **P1** by the CVX toolbox in MATLAB;
5: Calculate the power coefficient vector $\mathbf{w}_p^t$ and $\mathbf{w}_c^t$ by eigenvalue decomposition or Gaussian randomization method;
6: Set $\mathbf{w}_p^{(t+1)} = \mathbf{w}_p^t$, $\mathbf{w}_c^{(t+1)} = \mathbf{w}_c^t$, calculate $\mathbf{w}^{(t+1)}$;
7: **repeat**
8: Set $n \leftarrow 1$;
9: Calculate $\tilde{i}$ according to (67);
10: Set $\tilde{g}_{n,i}$ according to (68);
11: Update the index: $n \leftarrow n + 1$;
12: **until** $n > N_r$
13: Update the assignment matrix $\tilde{\mathbf{G}}$;
14: Set $\mathbf{G}^{(t+1)} = \tilde{\mathbf{G}}$;
15: Calculate the CRLB criterion $\mathrm{Tr}(\mathbf{J_u}^{-1}(\mathbf{w}_p^t, \mathbf{G}^t))$;
16: Update the index: $t \leftarrow t + 1$;
17: Optimize $\mathbf{w}_p$, $\mathbf{w}_c$ and $\mathbf{G}$ following a procedure similar to steps 3 through 16;
18: **until** $\mathrm{Tr}(\mathbf{J_u}^{-1})^{(t)} - \mathrm{Tr}(\mathbf{J_u}^{-1})^{(t-1)} \leq \epsilon$.
19: Record $\mathbf{w}_p^* = \mathbf{w}_p^t$, $\mathbf{w}_c^* = \mathbf{w}_c^t$, and $\mathbf{G}^* = \mathbf{G}^t$.

4.4. Computational complexity analysis

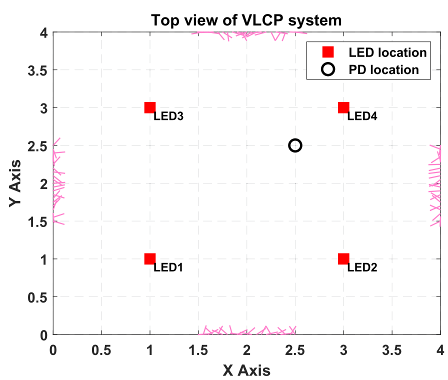
Assuming that the overall number of iterations is denoted by $\mathcal{I}$, the chosen of $\tilde{G}$ at each iteration needs $N_r N_{N_L}$ operations, thereby leading to a complexity of $O(N_r N_{N_L})$. This is because for an arbitrary RIS element, the calculation of $\tilde{i}$ based on (67) needs N_{N_L} operations at the worst

Optics EXPRESS

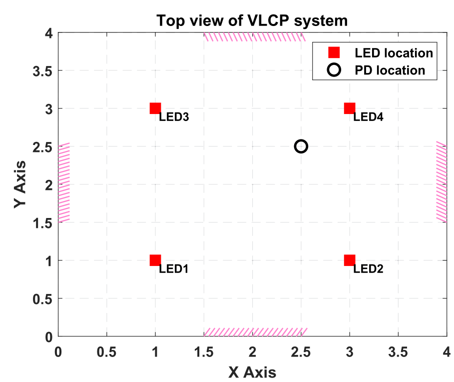
case, and meanwhile, there are N_r RIS element needed to be set based on (68) in our proposed RIS-aided VLCP system. Next, for the solution of the relaxed SDP problem, the CVX toolbox is employed and the worst-case computational complexity is $O((N_{N_L})^{4.5}\log(1/\delta_{\text{in}}))$, where δ_{in} is the accuracy of the interior point method [42]. Consequently, the overall computational complexity of the proposed algorithm can be expressed as $O(\mathcal{I}(N_r N_{N_L} + (N_{N_L})^{4.5}\log(1/\delta_{\text{in}})))$.

5. Numerical simulation results

In this section, some simulations are made to demonstrate the efficiency of our proposed algorithm in solving the proposed optimization problem. Meanwhile, the efficiency of RIS in improving the performance of VLCP is demonstrated according to our simulations, especially in the absence of LoS links. As shown in Fig. 3, we consider an indoor $4\times4\times3(\text{m})^3$ RIS-aided integrated multi-LED VLCP room with four LEDs and one PD. The locations of four LEDs are $(1, 1, 2.5)$m, $(3, 1, 2.5)$m, $(1, 3, 2.5)$m and $(3, 3, 2.5)$m, respectively. The location of the PD is set as $(2.5, 2.5, 0.5)$m in the simulation. The RIS is deployed on the four walls to increase the positioning and communication performance. The placement of blockers (users and non-users) follows a random uniform distribution. Users and non-users are modeled as cylinders with a 0.30-meter diameter and 1.65-meter height. The receiver is deemed blocked if the line from the LED to the PD intersects the cylinder. All results are averaged over 10,000 independent realizations.



(a)



(b)

Fig. 3. Top view of the RIS-aided VLCP system.

Figure 3(a) indicates that the RIS is in off-line status whereas Fig. 3(b) represents that the RIS is in on-line status. The pink short line represents the orientations of RIS elements in Fig. 3(a) and Fig. 3(b). The deployment of RIS is constrained in four rectangular regions and centered on the center point of the four walls. Ignoring the depth of walls, the center points of the four RISs are $(2, 0, 1.5)$m, $(0, 2, 1.5)$m, $(2, 4, 1.5)$m and $(4, 2, 1.5)$m and the area of RIS element is 0.01m^2. The FoV is assumed as 80^{o} and the incident angles of LoS links or RIS-reflected NLoS links can be verified within the FoV. All the other simulation parameters are summarized in Table 1 [14,32].

Figure 4 indicates the possible negative effect caused by RIS in the positioning process of VLCP system. In this figure, we assume that the power of LED is all allocated to the positioning process to observe the influence of RIS on the positioning accuracy. The range of noise variance values depicted in the figure aligns with the practical values described in the literature [3]. This alignment is based on the understanding that altering the noise variance would produce effects similar to those caused by adjusting the transmit power levels of the LEDs. It is observed that

Optics EXPRESS

Table 1. Simulation parameters

A	0.007	I_{DC}	2
$\Theta_{1/2}$	60^o	A_{re}	1cm^2
$T_{of}(\cdot)$	1	N_r	4×100
T_p	$0.1\mu s$	κ	0.53A/W
P_o	8(W)	P_e	16(W)
α	1.5	Ψ_{FoV}	80^o
δ	0.95	B_c	40MHz

the positioning performance improves as the signal-to-noise ratio (SNR) increases, indicated by a decrease in the CRLB. From the Figure, "CRLB LoS" means that the received positioning signals only come from the four LoS links, "CRLB NLoS" means that only RIS-reflected NLoS links provide the positioning signal (i.e., four LEDs are blocked), "CRLB Sum" means that the received positioning signals are provided by all LoS links and RIS-constructed NLoS links. By analyzing the Figure, we can obtain that the application of RIS deteriorates the positioning accuracy. This is since the position of receiver and the placement of RIS will influence the sign of partial derivative between the RIS-constructed and LoS-constructed derivation, and consequently, influence the value of (37). Consequently, we propose the LoS-dominance and RIS-secondary positioning scheme to avoid the negative positioning effect.

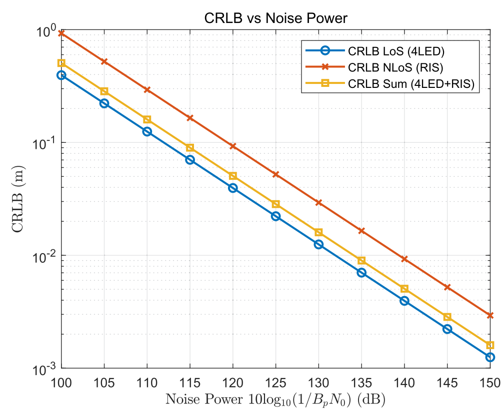


Fig. 4. CRLB versus noise power for RIS-assisted VLCP system with four LEDs and RIS.

Figure 5 demonstrates the efficiency of RIS on the positioning performance when one LoS link is absent. From the figure, assuming that the LoS link given by the LED 4 is blocked, the positioning performance is extremely deteriorated since that the LED 4 is closest to the receiver and plays an important role in the positioning process. Consequently, from the figure, we can observe that the RIS can play an active role in improving the positioning accuracy when one LoS link is absent.

Considering the constraint on the outage probability, assuming the noise power is constant, we make the simulations to investigate the relationship between the CRLB and the rate threshold given the outage probability. Figure 6 indicates the relationship between the CRLB and the rate threshold given the outage probability and the noise power. From the figure, we can observe that the application of RIS enhances positioning performance compared to the system using only LoS transmission. This is since that the employment of RIS is able to provide the NLoS channel gain to assist the communication, and therefore, relax the requirement of the communication power.

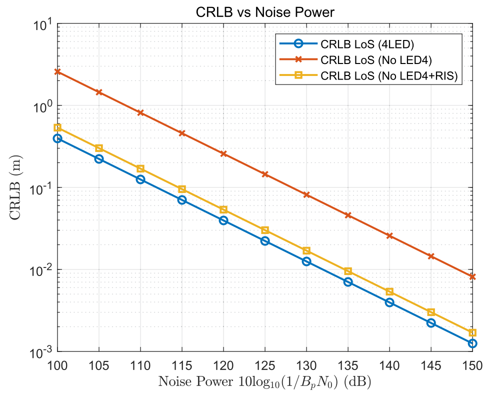


Fig. 5. CRLB versus noise power for RIS-assisted VLCP system with three LEDs and RIS.

Based on the reason, the positioning power can be allocated more to enhance the positioning performance. Meanwhile, the increase of the outage probability will enhance the positioning accuracy since the enhanced outage probability will relax the requirement of the communication, and therefore, allow less allocated power on the communication and more allocated positioning power to increase the CRLB.

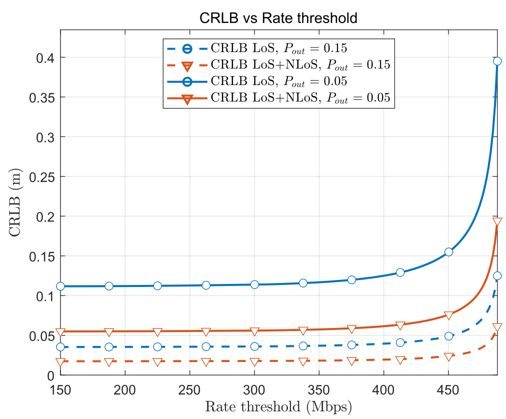


Fig. 6. CRLB versus rate threshold given the outage probability and the noise power $10\log_{10}(1/B_p N_0) = 130$(dB).

We choose the cumulative distribution function (CDF) of achievable data rate as the criterion of the communication performance. Figure 7 represents the CDF of the achievable rate given the outage probability with rate thresholds satisfied with $R_c = 200$Mbps. In the figure, "LoS" means that the channel gain is uniquely provided by the LoS channel while "LoS+NLoS" indicates that the transmission is provided by both LoS links and RIS-constructed NLoS links. From the figure, we can observe that the intensification of the outage probability will provide better communication performance since more power is allocated to the communication to satisfy the probability requirement. Meanwhile, we can obtain that the application of RIS can improve the communication performance obviously. This is since the considerable channel gain provided by RIS.

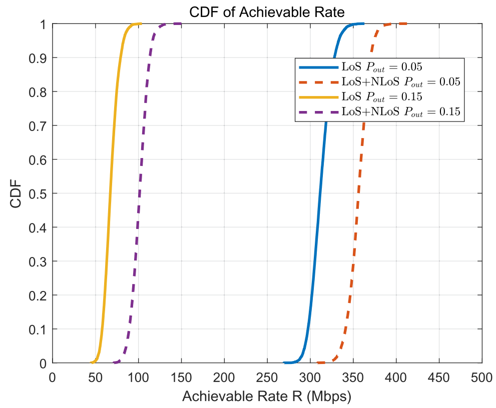


Fig. 7. CDF of the achievable rate given the outage probability with rate thresholds $R_c = 200$Mbps.

6. Conclusion

In this paper, we propose a RIS-aided positioning-centered multi-LED VLCP system to achieve the capability of VLC and VLP and relax the sensitivity to LoS links of visible light. We explain the mechanism of integrating communication and positioning and the timing of RIS operation within the VLCP system. The scheme is proposed to maximize the advantages of RIS in communication while avoiding adverse effects on positioning. Additionally, the RIS is utilized to supplement positioning when the LoS transmission is unavailable. The CRLB on the positioning result considering the RIS-reflected channel gain is derived and the maximum outage probability is considered as the constraint on the communication performance. In order to maximize the positioning accuracy, the RIS parameters and the transmission power are joint optimized to minimize the CRLB while satisfying the requirement of the maximum tolerate outage probability, RIS parameter acquirement and total transmit power constraints. To address the non-convex optimization problem, the AO algorithm, CVaR-based probability transform, SCA techniques and MRPC principle are proposed to optimize the RIS parameters and the transmission power. Simulation results represent that the application of RIS can efficiently improve the communication performance, relax the requirement of the communication power, and therefore, enhance the position performance. Meanwhile, from the simulation results, the efficiency of RIS in improving the positioning performance can be verified, especially when LoS links are absent. Consequently, the RIS can be considered as an effective option for the VLCP system in the upcoming era of the sixth generation.

7. Appendix

7.1 The CVaR-based transform for constraint in (47a)

For the distribution of the CSI error $\mathbf{e_h}$, we can obtain the expectation vector $\boldsymbol{\mu}$ and the covariance matrix $\boldsymbol{\Sigma}$. Assuming the CSI error is subject to the ambiguity set denoted by $\mathcal{P}$, and the probability distribution is represented by $\mathbb{P}$, the expectation vector and covariance matrix of CSI error in the ambiguity set are $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, respectively. Based on this, the constraint in (47a) can be written as

$$\begin{aligned}\inf_{\mathbb{P}\in\mathcal{P}} \Pr\{&\hat{\mathbf{h}}^\top(-\mathbf{W}_c)\hat{\mathbf{h}} + 2\hat{\mathbf{h}}^\top(-\mathbf{W}_c)\mathbf{e_h} \\ &+ \mathbf{e_h}^\top(-\mathbf{W}_c)\mathbf{e_h} + \varsigma \leq 0\} \geq 1 - P_{out}.\end{aligned} \tag{69}$$

Lemma 1 *[40,41] if there is a continuous function C which is quadratic or concave in $\boldsymbol{\sigma}$, i.e., $C(\boldsymbol{\sigma}) : \mathbb{R}^n \to \mathbb{R}$, the distributionally robust chance constraint can be expressed as its worst-case*

Optics EXPRESS

counterpart, given by

$$\begin{aligned} &\inf_{\mathbb{P}\in\mathcal{P}}\{\Pr_{\mathbb{P}}\{C(\boldsymbol{\sigma})\leq 0\}\}\geq 1-P_{out} \\ &\quad\Leftrightarrow \sup_{\mathbb{P}\in\mathcal{P}}\{\mathbb{P}-\mathrm{CVaR}\{C(\boldsymbol{\sigma});P_{out}\}\}\leq 0, \end{aligned} \tag{70}$$

where $\mathbb{P}-\mathrm{CVaR}\{C(\boldsymbol{\sigma});P_{out}\}$ *means the the Conditional Value-at-Risk of function* $C(\boldsymbol{\sigma})$ *under the probabilistic threshold constraint* P_{out} *when* $\boldsymbol{\sigma}$ *follows probability distribution* $\mathbb{P}$*, given by*

$$\begin{aligned} &\mathbb{P}-\mathrm{CVaR}\{C(\boldsymbol{\sigma});P_{out}\} \\ &\quad\triangleq \inf_{\lambda\in\mathcal{R}}\{\lambda+\frac{1}{P_{out}}\mathbb{E}\{(C(\boldsymbol{\sigma})-\lambda)^{+}\}\}, \end{aligned} \tag{71}$$

where λ *is the auxiliary variable, and* $(\cdot)^{+}\triangleq\max\{0,\cdot\}$.

Lemma 2 *[40,41] For the quadratic function* $C(\boldsymbol{\sigma})=\boldsymbol{\sigma}^{\top}\mathbf{A}\boldsymbol{\sigma}+\mathbf{b}^{\top}\boldsymbol{\sigma}+c$*, the CVaR in the worst case is the supremum of the function* $\mathbb{P}-\mathrm{CVaR}\{C(\boldsymbol{\sigma});P_{out}\}$ *characterized by*

$$\begin{aligned} \sup_{\mathbb{P}\in\mathcal{P}}\{\mathbb{P}-\mathrm{CVaR}\{C(\boldsymbol{\sigma});P_{out}\}\} &= \min_{\lambda,\mathbf{D}}\lambda+\frac{1}{P_{out}}\mathrm{Tr}(\mathbf{\Omega}\mathbf{D}) \\ s.t.\quad &\mathbf{D}\in\mathbb{S}^{n+1},\mathbf{D}\succeq\mathbf{0}, \\ &\mathbf{D}-\begin{bmatrix}\mathbf{A} & \frac{1}{2}\mathbf{b}\\ \frac{1}{2}\mathbf{b}^{\top} & c-\lambda\end{bmatrix}\succeq\mathbf{0}, \\ &\mathbf{\Omega}=\begin{bmatrix}\Sigma+\boldsymbol{\mu}\boldsymbol{\mu}^{\mathrm{T}} & \boldsymbol{\mu}\\ \boldsymbol{\mu}^{\mathrm{T}} & \mathbf{1}\end{bmatrix}, \end{aligned} \tag{72}$$

where $\mathbf{D}$ *is the auxiliary variable,* $\boldsymbol{\mu}$ *and* Σ *are the expectation and covariance matrix of random variable* $\boldsymbol{\sigma}$.

Combined with **Lemma** 1 and **Lemma** 2, let $\mathbf{A}=-\mathbf{W}_c$, $\mathbf{b}=2\hat{\mathbf{h}}^{\top}(-\mathbf{W}_c)\mathbf{e_h}$, $c=\varsigma$, the constraint in (47a) can be rewritten as

$$\lambda+\frac{1}{P_{\mathrm{out}}}\mathrm{Tr}(\mathbf{\Omega}\mathbf{D})\leq 0 \tag{73a}$$

$$\mathbf{D}-\begin{bmatrix}-\mathbf{W}_c & -\mathbf{W}_c^{\mathrm{T}}\hat{\mathbf{h}}\\ -\hat{\mathbf{h}}^{\mathrm{T}}\mathbf{W}_c & -\hat{\mathbf{h}}^{\mathrm{T}}\mathbf{W}_c\hat{\mathbf{h}}+\varsigma-\lambda\end{bmatrix}\succeq\mathbf{0} \tag{73b}$$

$$\mathbf{D}\succeq\mathbf{0},\mathbf{D}\in\mathbb{S}^4 \tag{73c}$$

where $\mathbf{\Omega}$ is the auxiliary variable expressed as

$$\mathbf{\Omega}=\begin{bmatrix}\Sigma+\boldsymbol{\mu}\boldsymbol{\mu}^{\mathrm{T}} & \boldsymbol{\mu}\\ \boldsymbol{\mu}^{\mathrm{T}} & \mathbf{1}\end{bmatrix}, \tag{74}$$

and here, Σ and $\boldsymbol{\mu}$ are the covariance matrix and expectation of random variable $\mathbf{e_h}$. This completes the proof of the CVaR-based transform for constraint in (47a).

Funding. National Natural Science Foundation of China (NSFC) (62071489); National Key Research and Development Program of China (2022YFB2802804).

Acknowledgment. The authors wish to thank the anonymous reviewers for their valuable suggestions. This study was supported by the National Natural Science Foundation of China (NSFC) under grant 62071489, the National Natural Science Foundation of China (NSFC) under grant 62331024, and the National Key Research and Development Program of China under grant (2022YFB2802804).

Optics EXPRESS

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

1. A. Jovicic, J. Li, and T. Richardson, "Visible light communication: opportunities, challenges and the path to market," IEEE Commun. Mag. **51**(12), 26–32 (2013).
2. C. Zeyu, "6g, lifi and wifi wireless systems: Challenges, development and prospects," in *2021 18th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP)*, (IEEE, 2021), pp. 322–325.
3. X. Zhang, J. Duan, Y. Fu, *et al.*, "Theoretical accuracy analysis of indoor visible light communication positioning system based on received signal strength indicator," J. Lightwave Technol. **32**(21), 4180–4186 (2014).
4. N. E. Klepeis, W. C. Nelson, W. R. Ott, *et al.*, "The national human activity pattern survey (nhaps): a resource for assessing exposure to environmental pollutants," J. Expo. Sci. Environ. Epidemiol. **11**(3), 231–252 (2001).
5. M. Obeed, A. M. Salhab, M.-S. Alouini, *et al.*, "On optimizing vlc networks for downlink multi-user transmission: A survey," IEEE Commun. Surv. Tutorials **21**(3), 2947–2976 (2019).
6. M. A. Arfaoui, M. D. Soltani, I. Tavakkolnia, *et al.*, "Measurements-based channel models for indoor lifi systems," IEEE Trans. Wireless Commun. **20**(2), 827–842 (2021).
7. S. Research, "6g: The next hyper connected experience for all," (2020). Available online: https://research.samsung.com/next-generation-communications (accessed on 24 July 2022).
8. C. Amini, A. Taherpour, T. Khattab, *et al.*, "Theoretical accuracy analysis of indoor visible light communication positioning system based on time-of-arrival," in *2016 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, (IEEE, 2016), pp. 1–5.
9. T.-H. Do and M. Yoo, "Tdoa-based indoor positioning using visible light," Photonics Netw. Commun. **27**(2), 80–88 (2014).
10. K. Panta and J. Armstrong, "Indoor localisation using white leds," Electron. Lett. **48**(4), 228–230 (2012).
11. P. Du, S. Zhang, W.-D. Zhong, *et al.*, "Real-time indoor positioning system for a smart workshop using white leds and a phase-difference-of-arrival approach," Opt. Eng. **58**(08), 084112 (2019).
12. S.-H. Yang, H.-S. Kim, Y.-H. Son, *et al.*, "Three-dimensional visible light indoor localization using aoa and rss with multiple optical receivers," J. Lightwave Technol. **32**(14), 2480–2485 (2014).
13. P. Du, S. Zhang, C. Chen, *et al.*, "Experimental demonstration of 3d visible light positioning using received signal strength with low-complexity trilateration assisted by deep learning technique," IEEE Access **7**, 93986–93997 (2019).
14. S. Ma, R. Yang, B. Li, *et al.*, "Optimal power allocation for integrated visible light positioning and communication system with a single led-lamp," IEEE Trans. Commun. **70**(10), 6734–6747 (2022).
15. S. Ma, R. Yang, C. Du, *et al.*, "Robust power allocation for integrated visible light positioning and communication networks," IEEE Trans. Commun. **71**(8), 4764–4777 (2023).
16. S. Ma, S. Cao, H. Li, *et al.*, "Waveform design and optimization for integrated visible light positioning and communication," IEEE Trans. Commun. **71**(9), 5392–5407 (2023).
17. H. Yang, W.-D. Zhong, C. Chen, *et al.*, "Qos-driven optimized design-based integrated visible light communication and positioning for indoor iot networks," IEEE Internet Things J. **7**(1), 269–283 (2020).
18. H. Yang, W.-D. Zhong, C. Chen, *et al.*, "Coordinated resource allocation-based integrated visible light communication and positioning systems for indoor iot," IEEE Trans. Wireless Commun. **19**(7), 4671–4684 (2020).
19. H. Yang, S. Zhang, A. Alphones, *et al.*, "An advanced integrated visible light communication and localization system," IEEE Trans. Commun. (2023).
20. S. Ma, H. Sheng, J. Sun, *et al.*, "Feasibility conditions for mobile lifi," IEEE Trans. Wireless Commun. **23**(7), 7911–7923 (2024).
21. S. Sun, W. Mei, F. Yang, *et al.*, "Optical intelligent reflecting surface assisted mimo vlc: Channel modeling and capacity characterization," IEEE Trans. Wireless Commun. (2023).
22. S. Sun, N. An, F. Yang, *et al.*, "Capacity characterization analysis of optical intelligent reflecting surface assisted miso vlc," IEEE Internet Things J. **11**(3), 4801–4814 (2024).
23. S. Aboagye, T. M. Ngatched, O. A. Dobre, *et al.*, "Intelligent reflecting surface-aided indoor visible light communication systems," IEEE Commun. Lett. **25**(12), 3913–3917 (2021).
24. S. Sun, F. Yang, J. Song, *et al.*, "Joint resource management for intelligent reflecting surface–aided visible light communications," IEEE Trans. Wireless Commun. **21**(8), 6508–6522 (2022).
25. B. Cao, M. Chen, Z. Yang, *et al.*, "Reflecting the light: Energy efficient visible light communication with reconfigurable intelligent surface," in *2020 IEEE 92nd Vehicular Technology Conference (VTC2020-Fall)*, (IEEE, 2020), pp. 1–5.
26. L. Qian, X. Chi, L. Zhao, *et al.*, "Secure visible light communications via intelligent reflecting surfaces," in *ICC 2021-IEEE International Conference on Communications*, (IEEE, 2021), pp. 1–6.
27. S. Sun, F. Yang, J. Song, *et al.*, "Optimization on multiuser physical layer security of intelligent reflecting surface-aided vlc," IEEE Wireless Commun. Lett. **11**(7), 1344–1348 (2022).
28. Q. Zhang, F. Yang, Z. Liu, *et al.*, "Hardware implementation of a mirror array-based optical intelligent reflecting surface for vlc: prototype and experimental results," Opt. Express **32**(11), 19252–19264 (2024).

Optics EXPRESS

29. I. Iddrisu and S. Gezici, "Visible light positioning with intelligent reflecting surfaces under mismatched orientations," arXiv (2024).
30. S. Xu, Y. Wu, L. Zhang, *et al.*, "Simultaneous position and orientation estimation for reconfigurable intelligent surfaces-assisted visible light communications," IEEE Transactions on Vehicular Technology (2024).
31. F. Kokdogan and S. Gezici, "Intelligent reflecting surfaces for visible light positioning based on received power measurements," IEEE Transactions on Vehicular Technology (2024).
32. R. Yang, G. Zhang, S. Ma, *et al.*, "Robust power allocation for multi-led integrated visible light positioning and communication," J. Electron. Inf. Technol. **46**(4), 1186–1195 (2024).
33. T. Komine and M. Nakagawa, "Fundamental analysis for visible-light communication system using led lights," IEEE Trans. Consumer Electron. **50**(1), 100–107 (2004).
34. M. Najafi, B. Schmauss, and R. Schober, "Intelligent reflecting surfaces for free space optical communication systems," IEEE Trans. Commun. **69**(9), 6134–6151 (2021).
35. S. Sun, F. Yang, W. Mei, *et al.*, "Optical irs for visible light communication: Modeling, design, and open issues," arXiv (2024).
36. V. Lottici, A. D'Andrea, and U. Mengali, "Channel estimation for ultra-wideband communications," IEEE J. Select. Areas Commun. **20**(9), 1638–1645 (2002).
37. J.-B. Wang, Q.-S. Hu, J. Wang, *et al.*, "Tight bounds on channel capacity for dimmable visible light communications," J. Lightwave Technol. **31**(23), 3771–3779 (2013).
38. M. F. Keskin, A. D. Sezer, and S. Gezici, "Optimal and robust power allocation for visible light positioning systems under illumination constraints," IEEE Trans. Commun. **67**(1), 527–542 (2019).
39. H. V. Poor, *An introduction to signal detection and estimation* (Springer Science & Business Media, 2013).
40. S. Zymler, D. Kuhn, and B. Rustem, "Distributionally robust joint chance constraints with second-order moment information," Math. Program. **137**(1-2), 167–198 (2013).
41. Y. Zhang, B. Li, F. Gao, *et al.*, "A robust design for ultra reliable ambient backscatter communication systems," IEEE Internet Things J. **6**(5), 8989–8999 (2019).
42. Z.-q. Luo, W.-k. Ma, A. M.-c. So, *et al.*, "Semidefinite relaxation of quadratic optimization problems," IEEE Signal Process. Mag. **27**(3), 20–34 (2010).
43. M. Grant and S. Boyd, "Cvx: Matlab software for disciplined convex programming, version 2.1," (2014), available online: http://cvxr.com/cvx.